COMP90042
Natural Language Processing

Lecture 12: Large Language Models (LLM)

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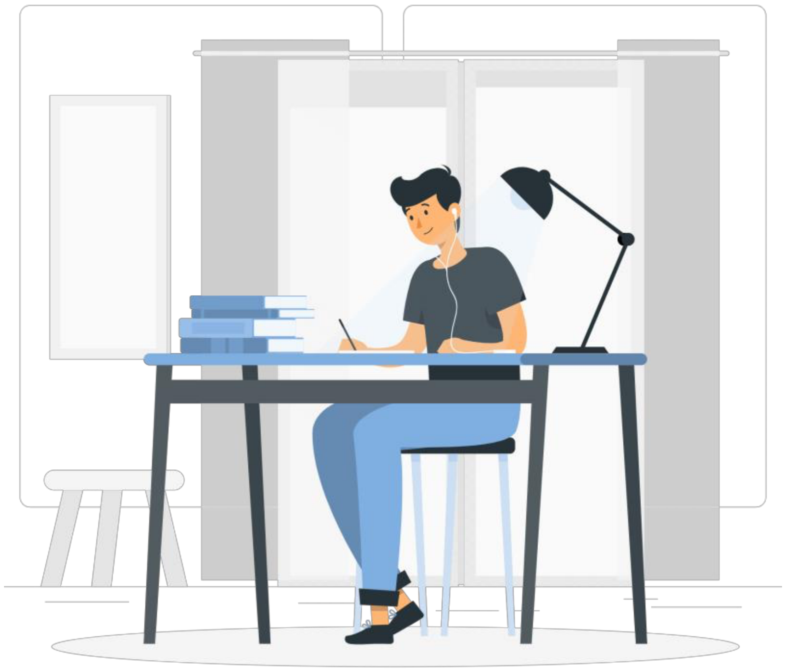
Semester 1, 2026
School of Computing and Information Systems
University of Melbourne

/ Project Assignment Overview

Project Specification will be released in Week 7

(Group Assignment: 3 group members)

* Some groups may have 2 group members, depending on the total number of students enrolled



Project *(Implementation + Report)*

Peer Review

/ Lecture Plan

Understanding Large Language Models

1. The Large Language Model
2. Transformer: Main Points and Decoder
3. Generative Pretrained Transformers (GPT) and LLM capabilities
    1. GPT-1: Unsupervised Pretraining + Supervised Fine-Tuning
    2. GPT-2: Scaling Up and Zero-shot Capabilities
    3. GPT-3: Few-shot and In-Context Learning
    4. ChatGPT: Reinforcement Learning from Human Feedback (RLHF)

I love natural language ____________

Language models predict the next word (or "token") in a sequence by calculating probabilities based on training data

p(next word | previous words)

I love natural language ________

Large Language models predict the next word (or "token") in a sequence by calculating probabilities based on **massive** training data

+ **massive** parameters (internal configuration variables)
+ **massive** computational power required to create them

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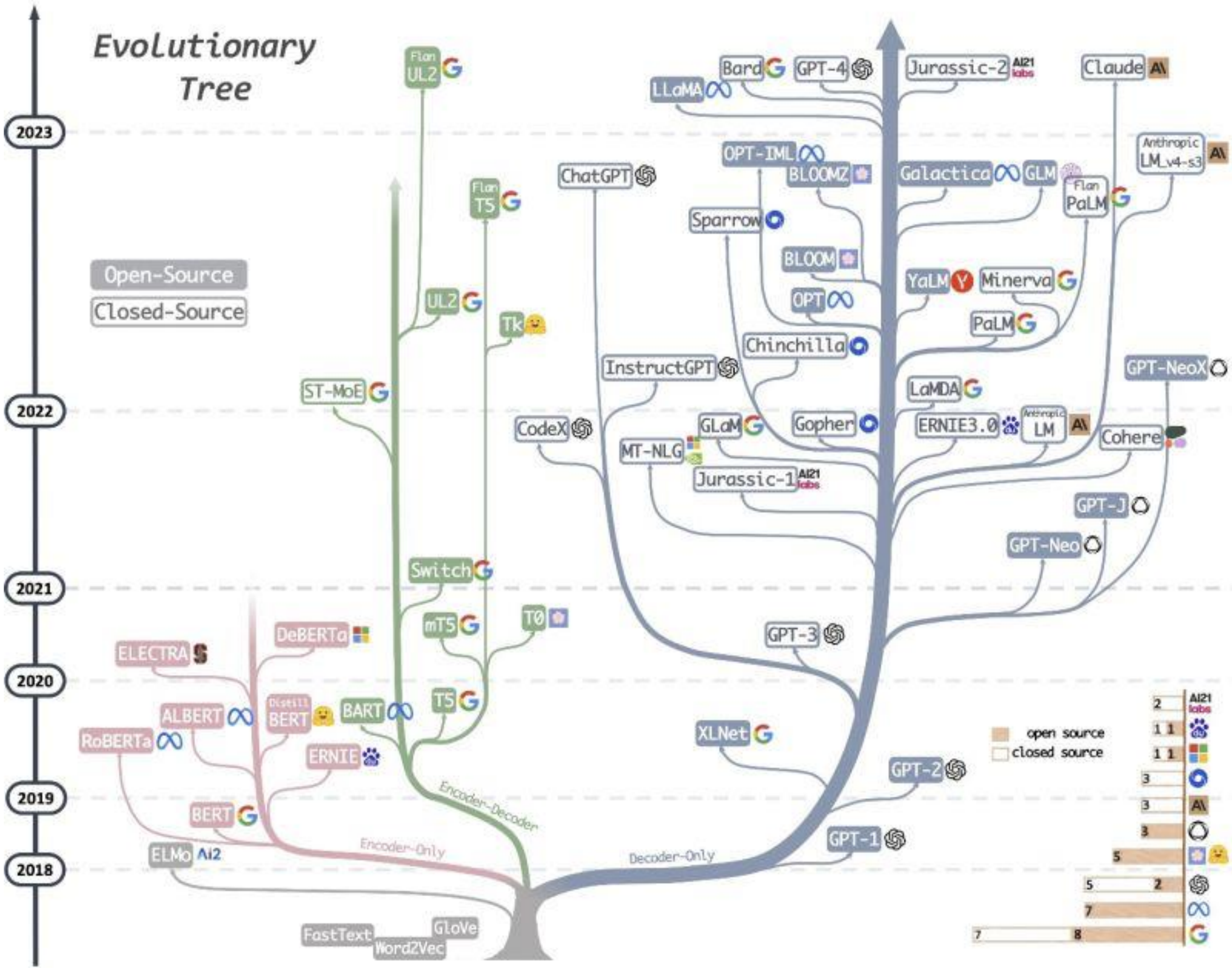
+ **massive** computational power required to create them

Large Language Models



1 Large Language Models

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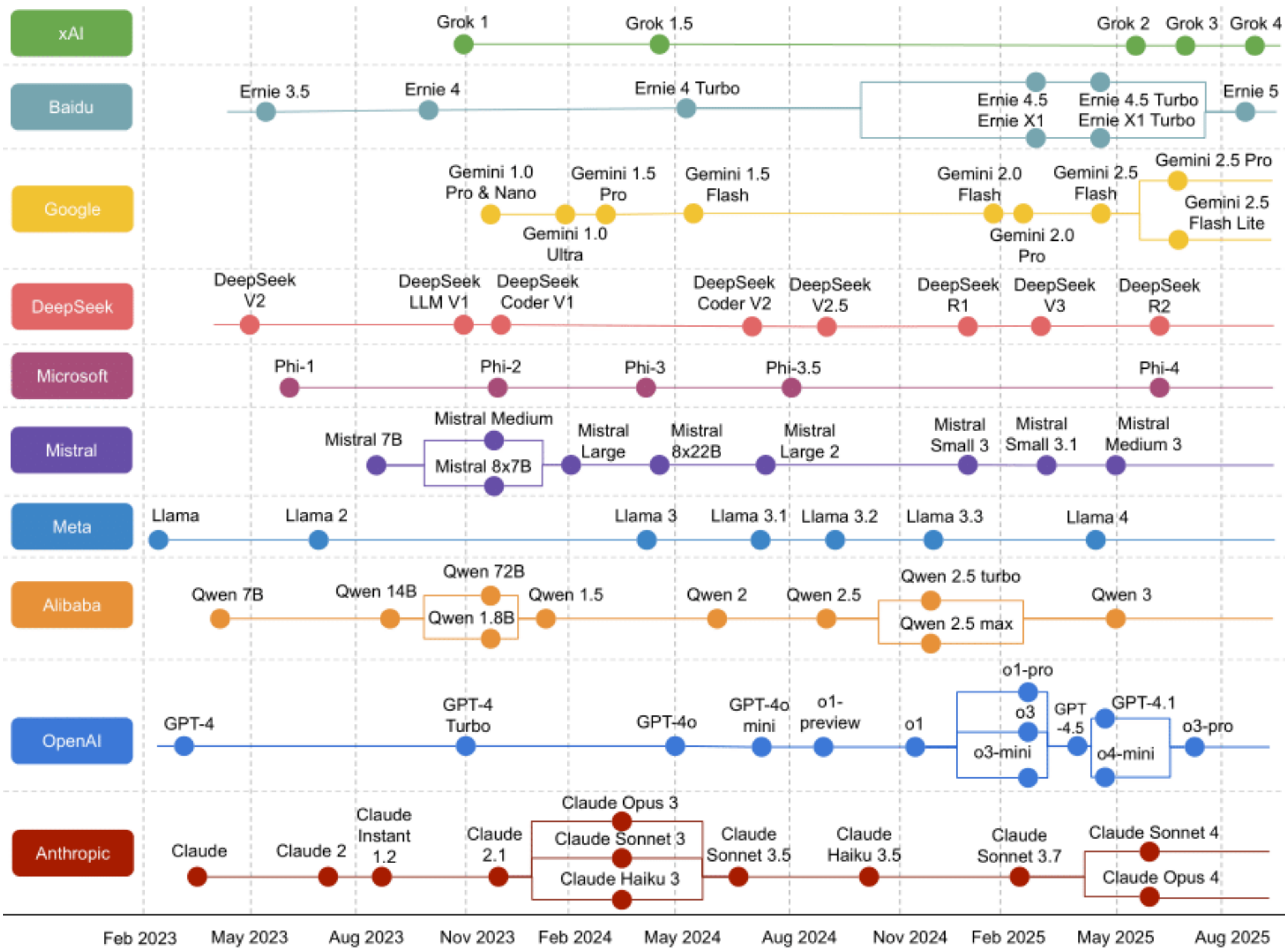

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Understanding Large Language Models

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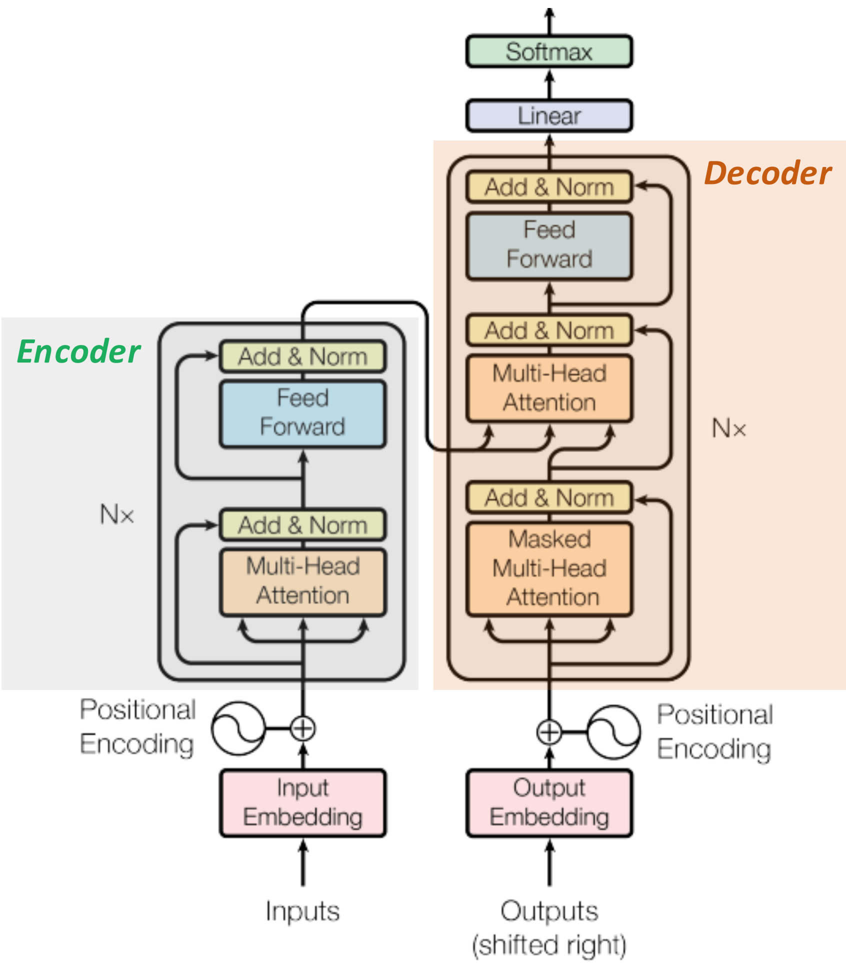
4. **Scaling Laws and LLAMA**

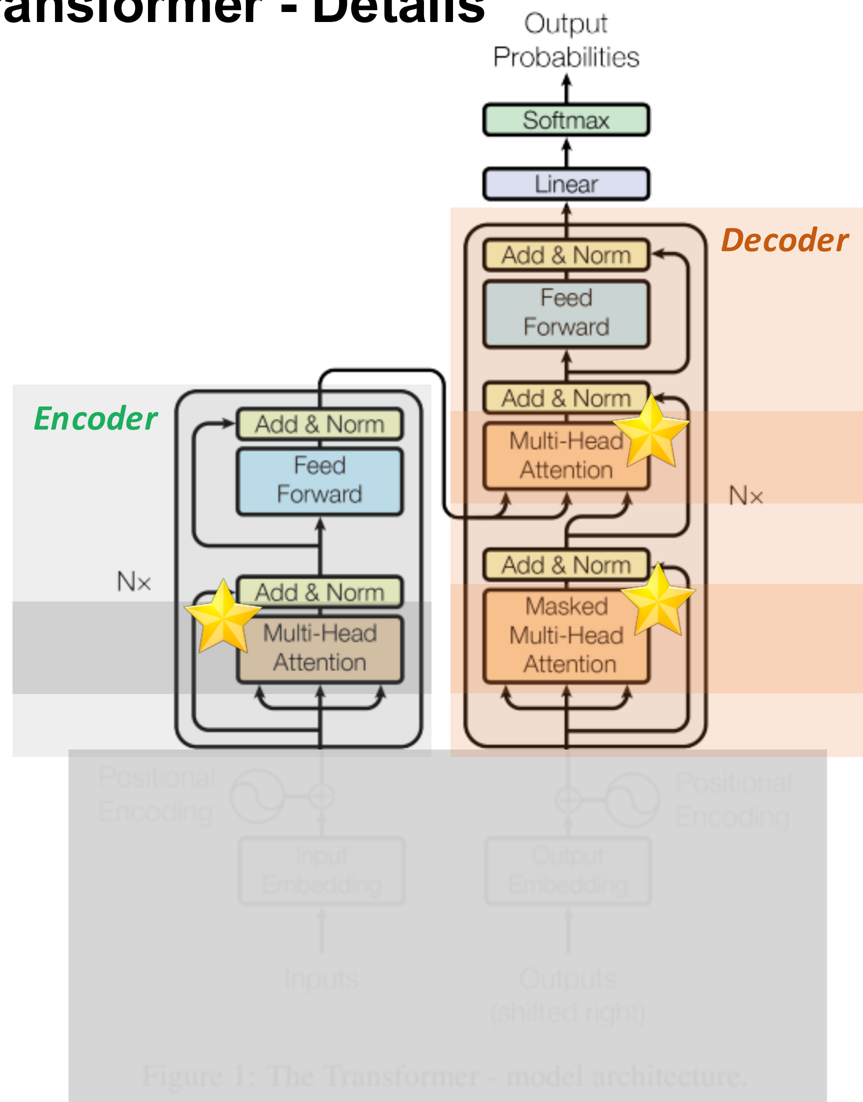
Transformer: Main Points and Decoder

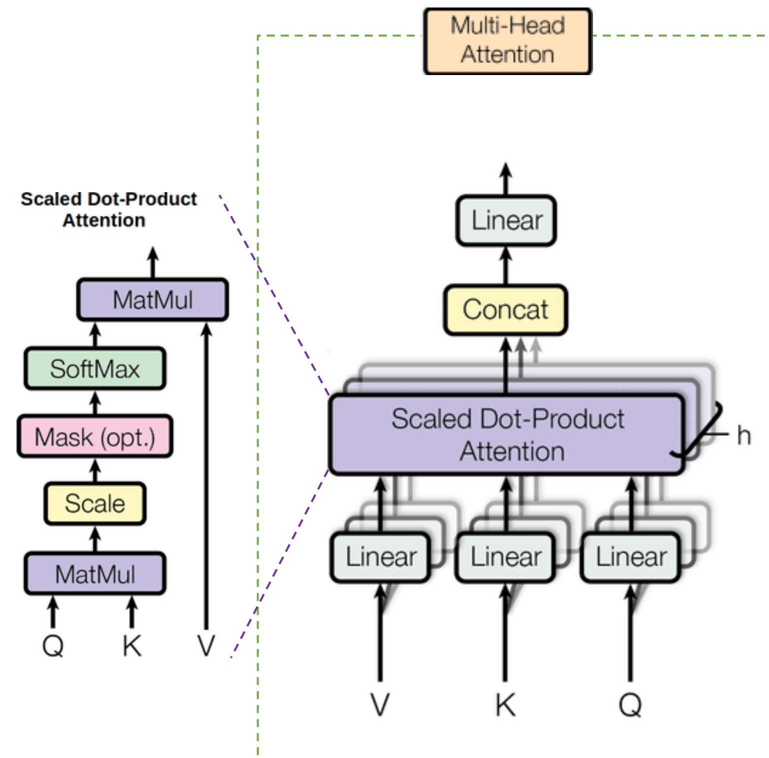


But Language Model Does not require ALL Components!

They focus on the context – so next word prediction only!

Transformer - Details





Text & Position Embedding

x_1 x_2

Hello ***Australia***

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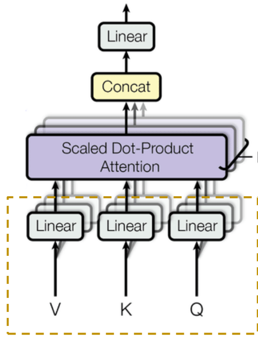
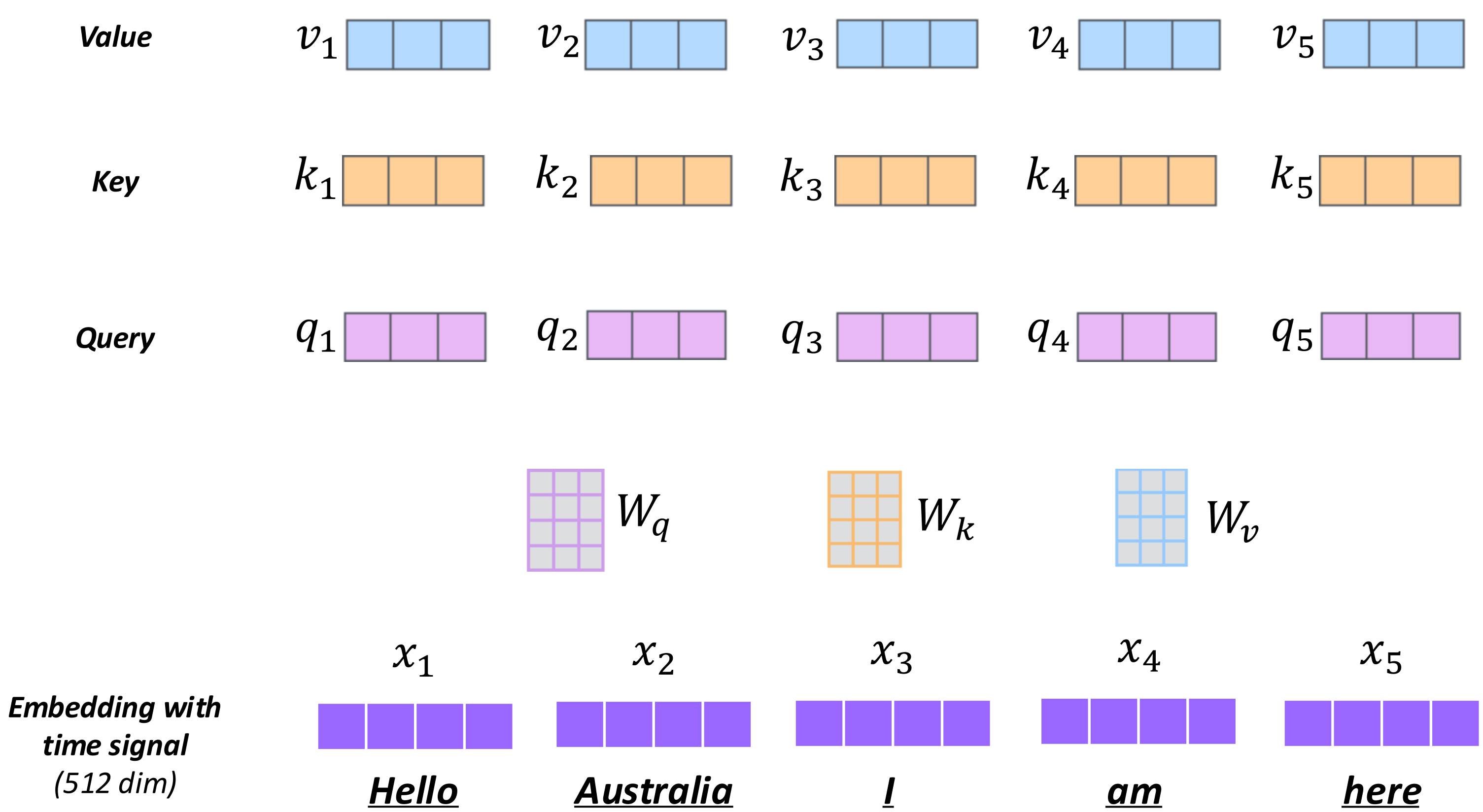
Transformer – Multi-head Self Attention

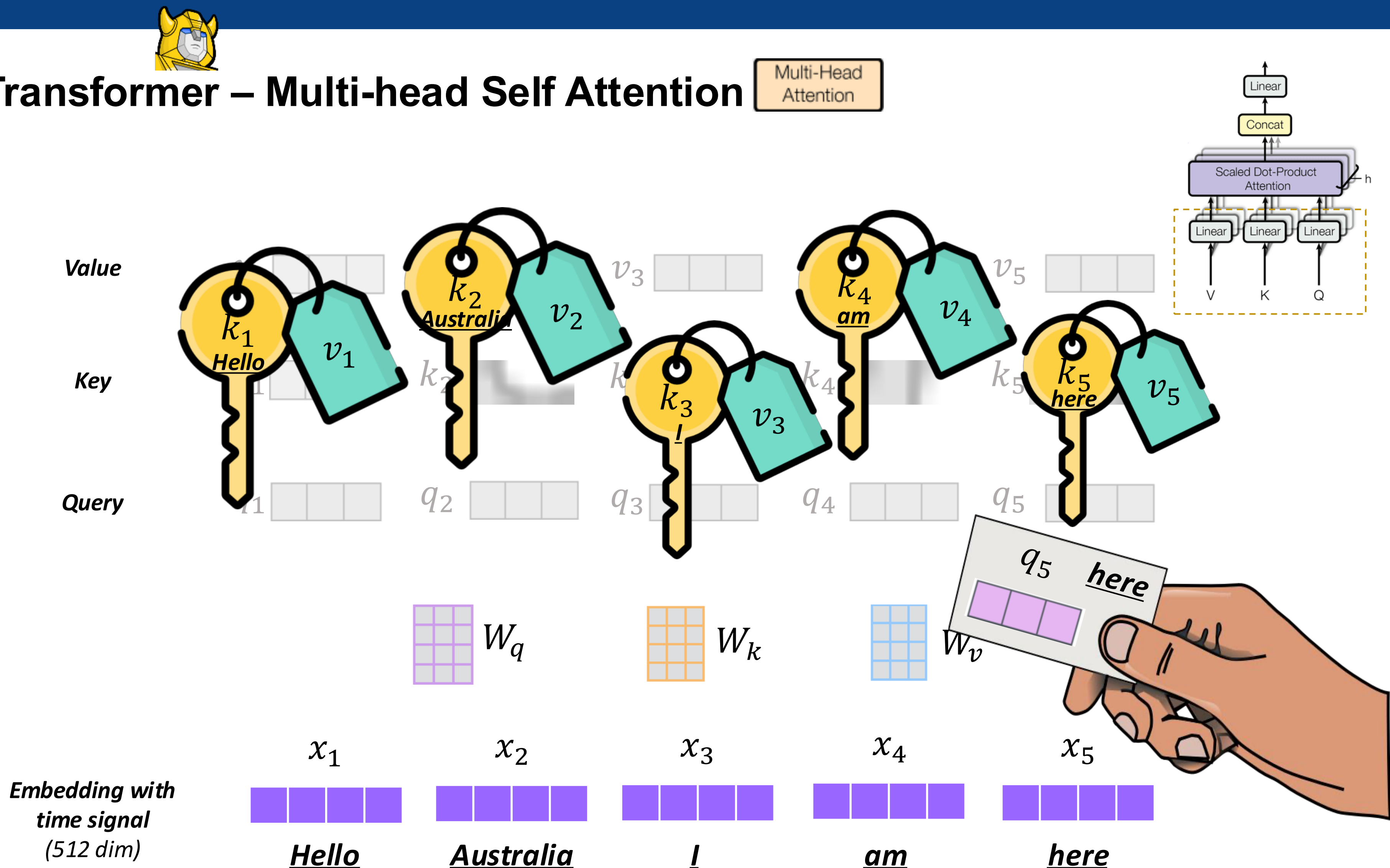
Multi-Head Attention





Dimensionality of q,k,v = 64

Transformer – Multi-head Self Attention



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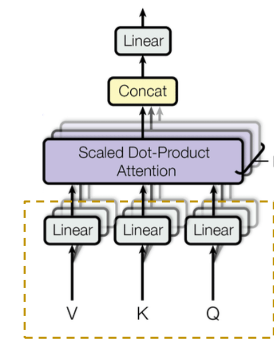
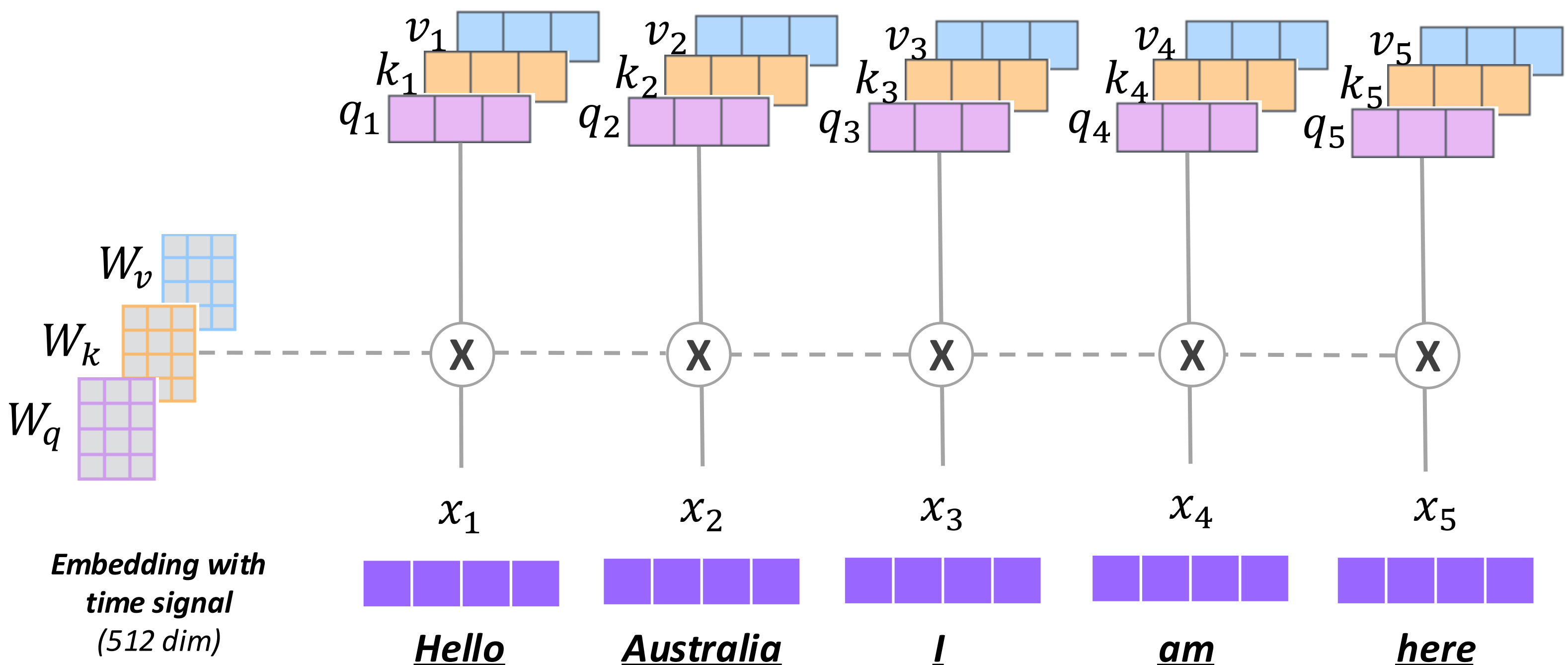
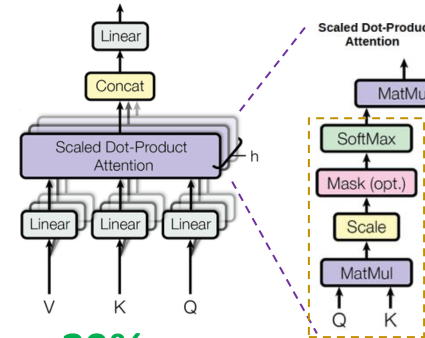
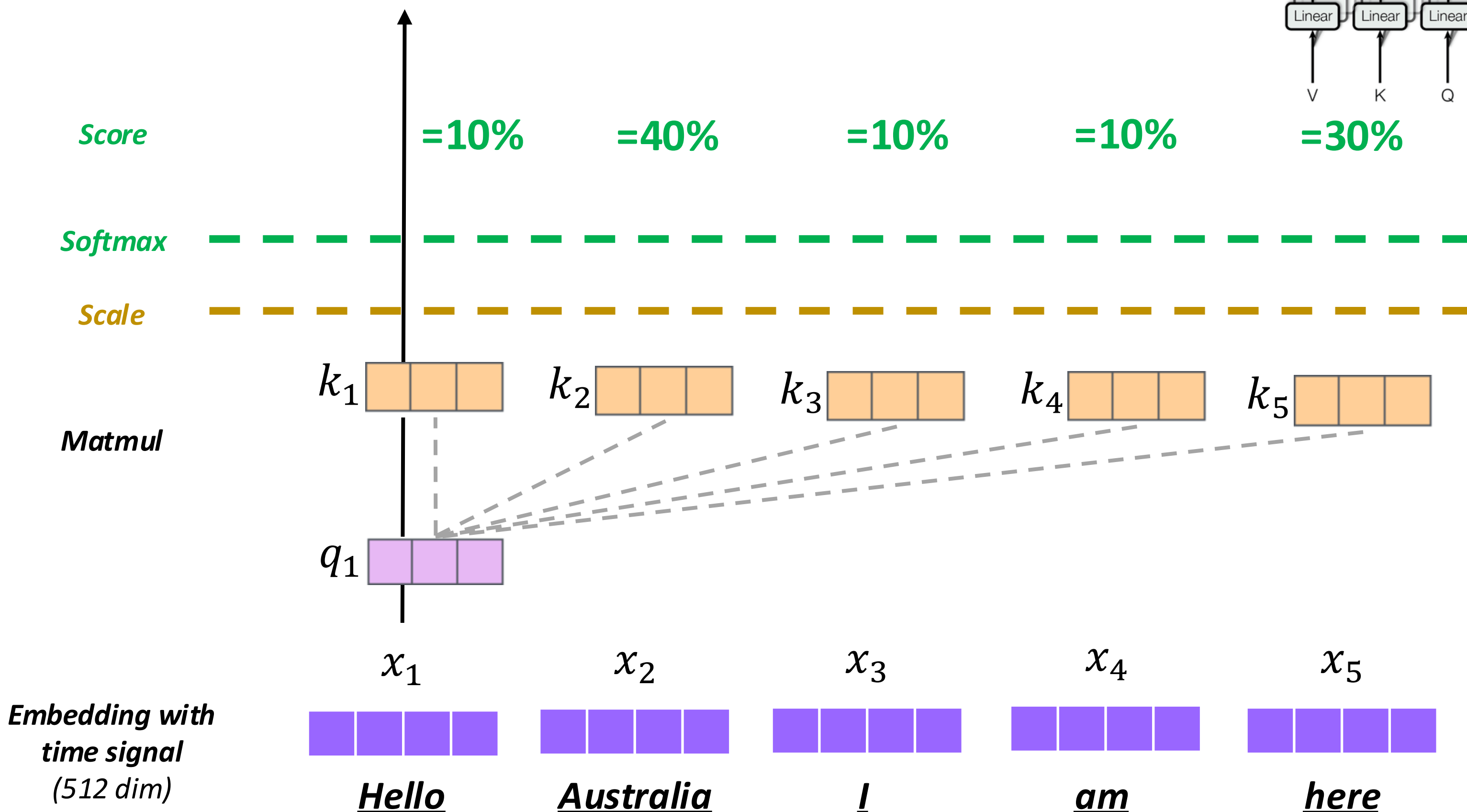




Transformer – Multi-head Self Attention

Multi-Head Attention

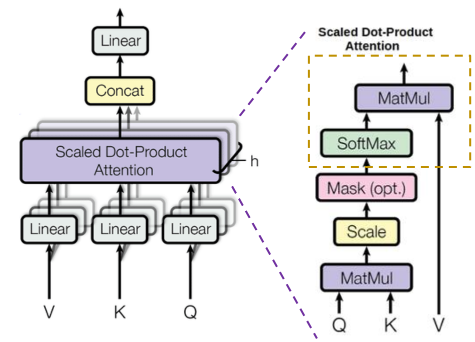


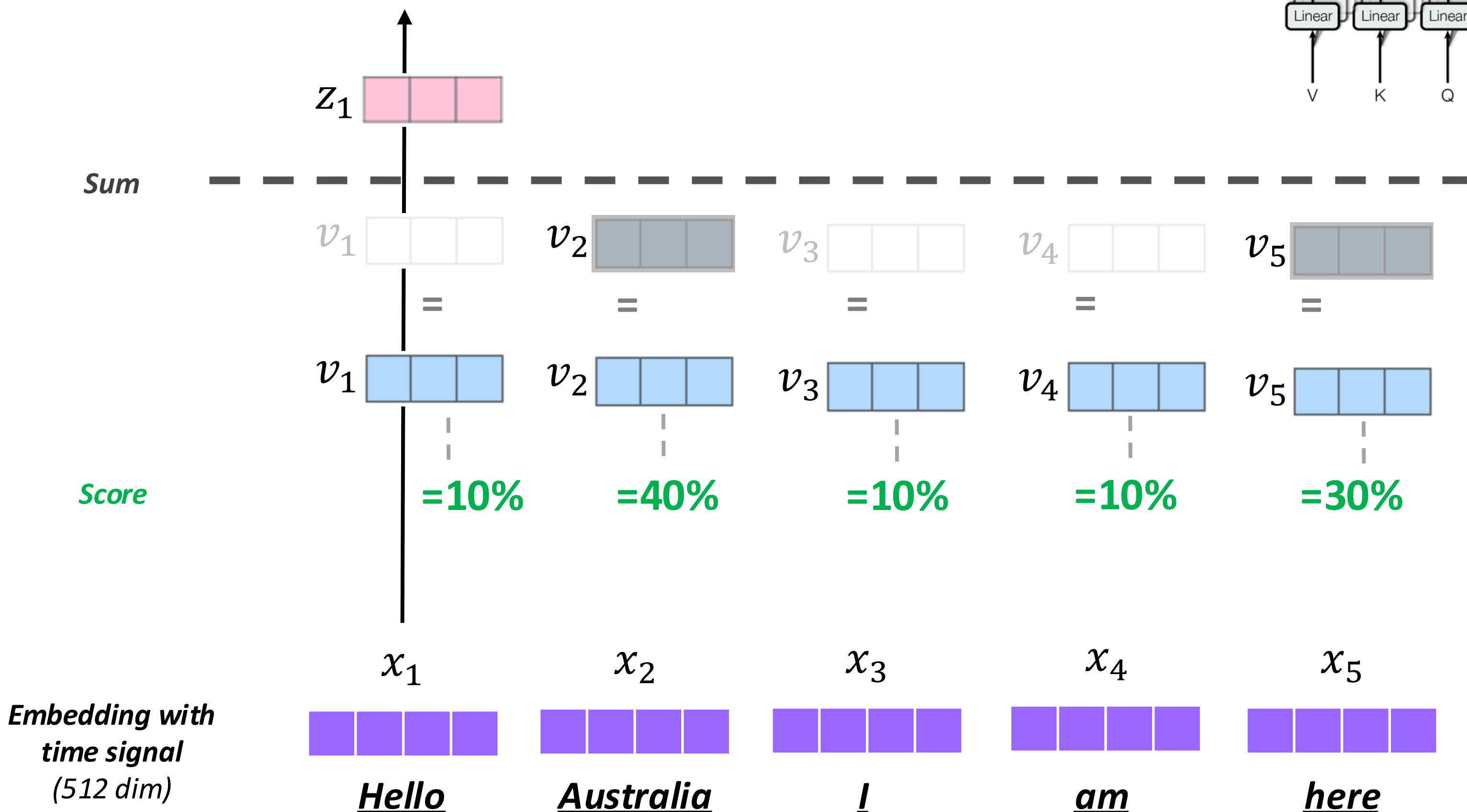


2 Transformer: Main Points and Decoder

Transformer – Multi-head Self Attention

Multi-Head Attention





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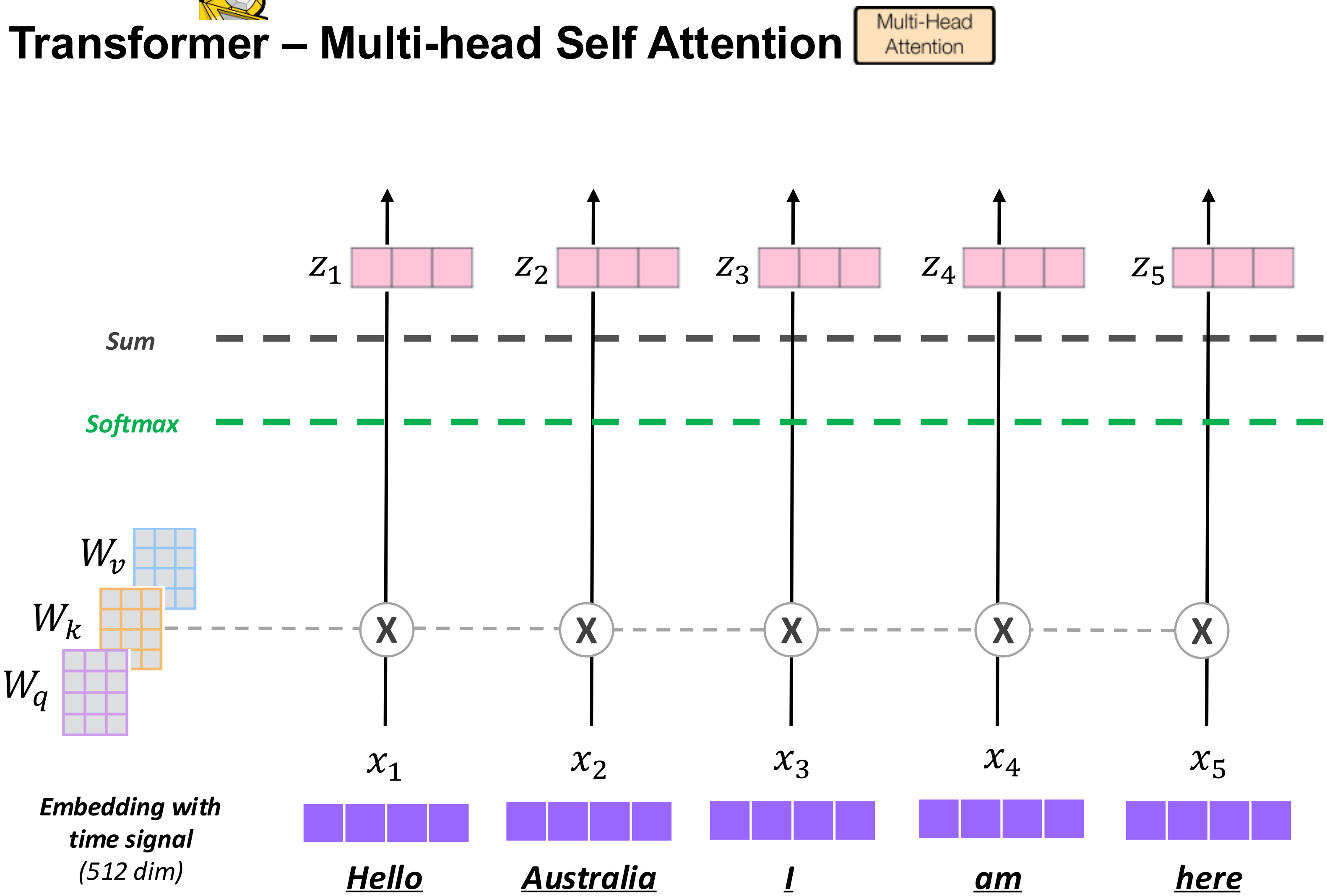
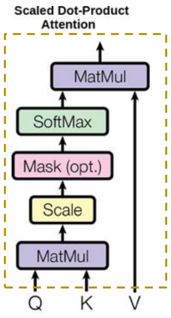
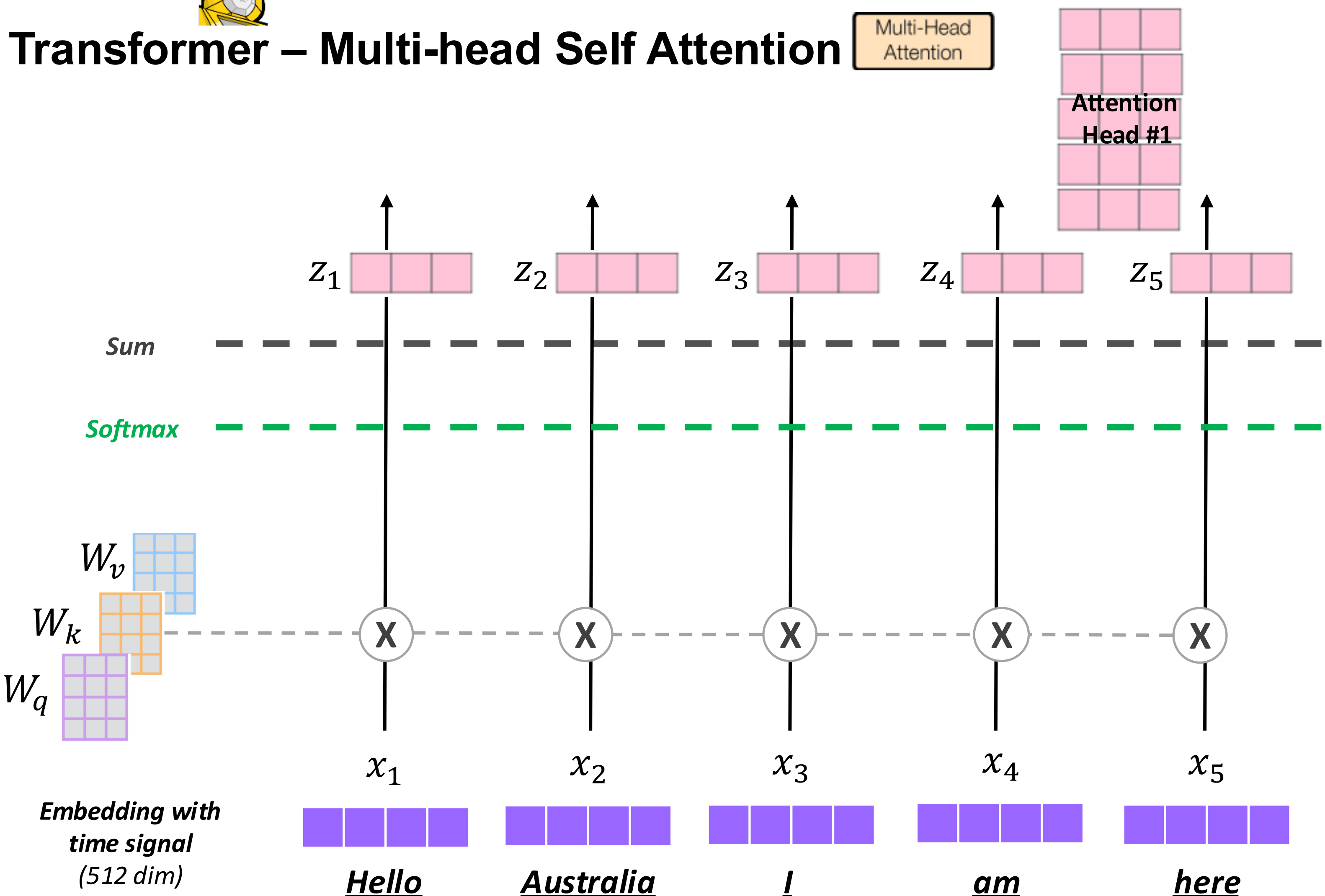
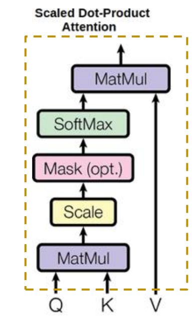




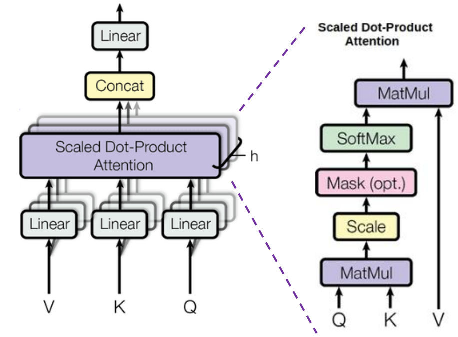
Transformer – Multi-head Self Attention





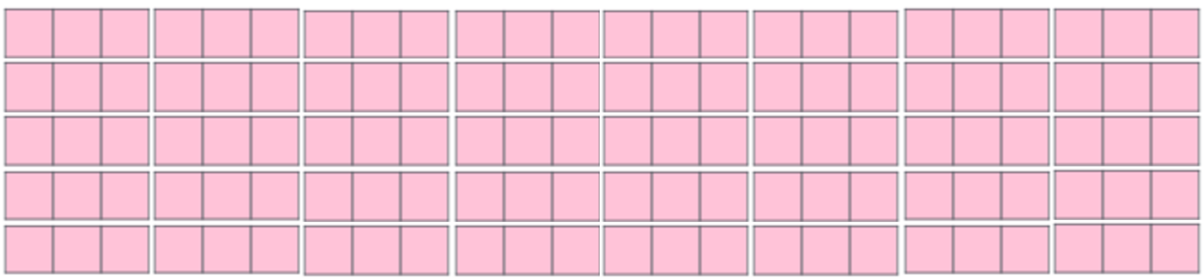
Transformer – Multi-head Self Attention

Multi-Head Attention



1) ***<u>Concatenate</u> all attention heads (****num_heads = 8****)***

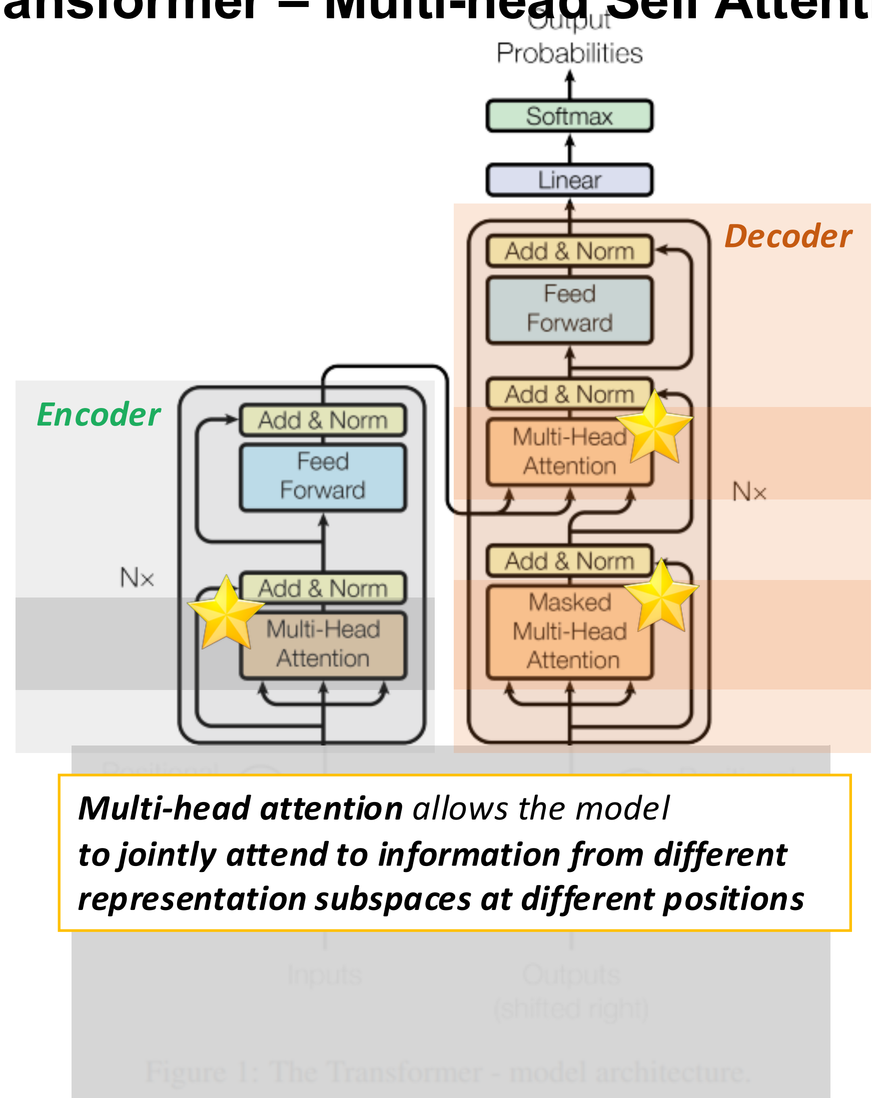
Attention Head **#1**	Attention Head **#2**	Attention Head **#3**	Attention Head **#4**	Attention Head **#5**	Attention Head **#6**	Attention Head **#7**	Attention Head **#8**



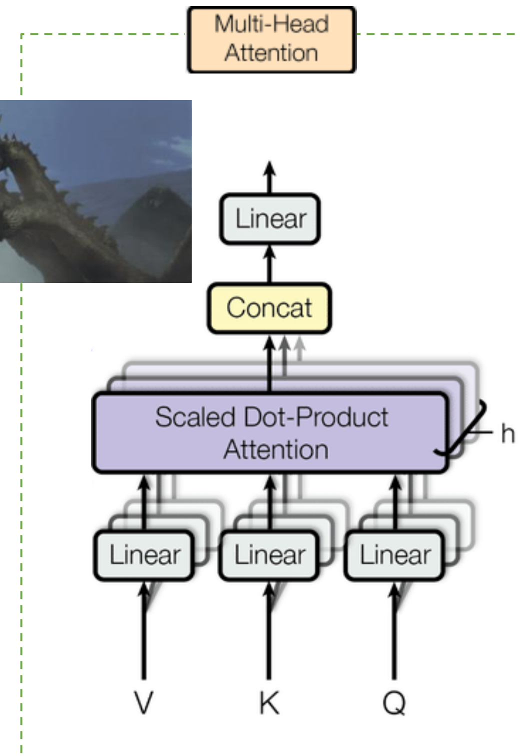
2) (Linear Layer) <u>Multiply</u> with a weight matrix *W that was trained jointly with the model*

3) The result would be Z *matrix that captures information from all the attention heads.*

Transformer – Multi-head Self Attention



Multi-head attention *allows the model*
to jointly attend to information from different representation subspaces at different positions



Text & Position Embedding

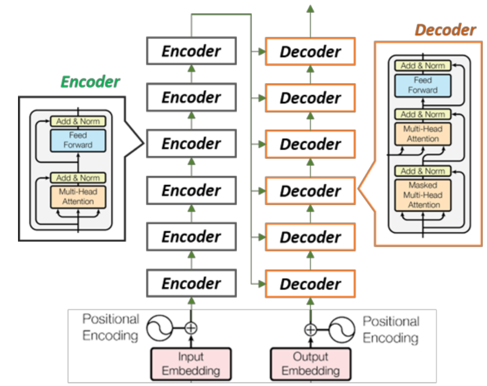
x_1 x_2

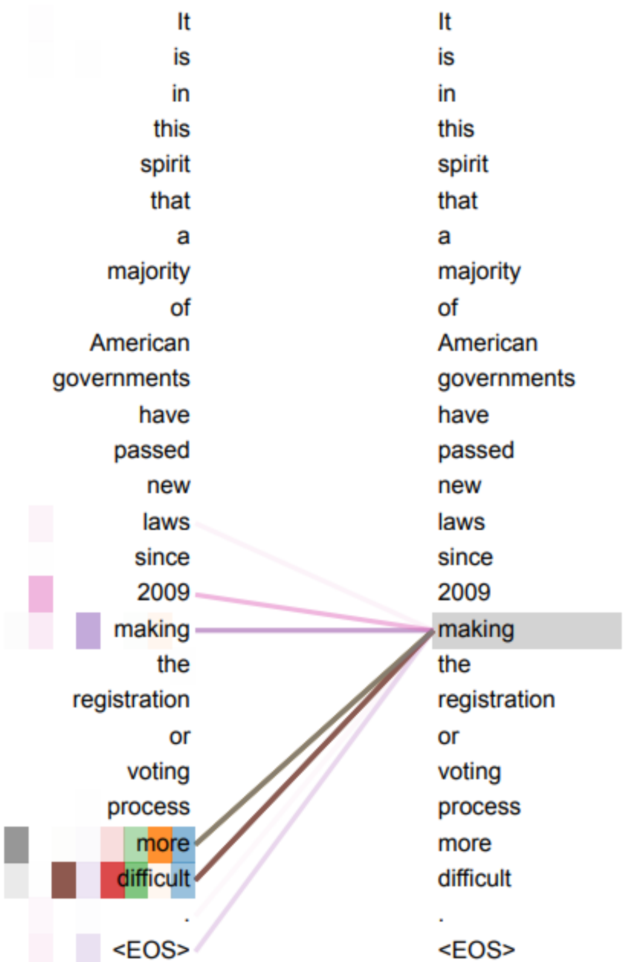
<u>Hello</u> ***<u>Australia</u>***

Transformer – Multi-head Self Attention

Multi-Head Attention

From the original **Transformer Paper (Vaswani et al., 2017)**





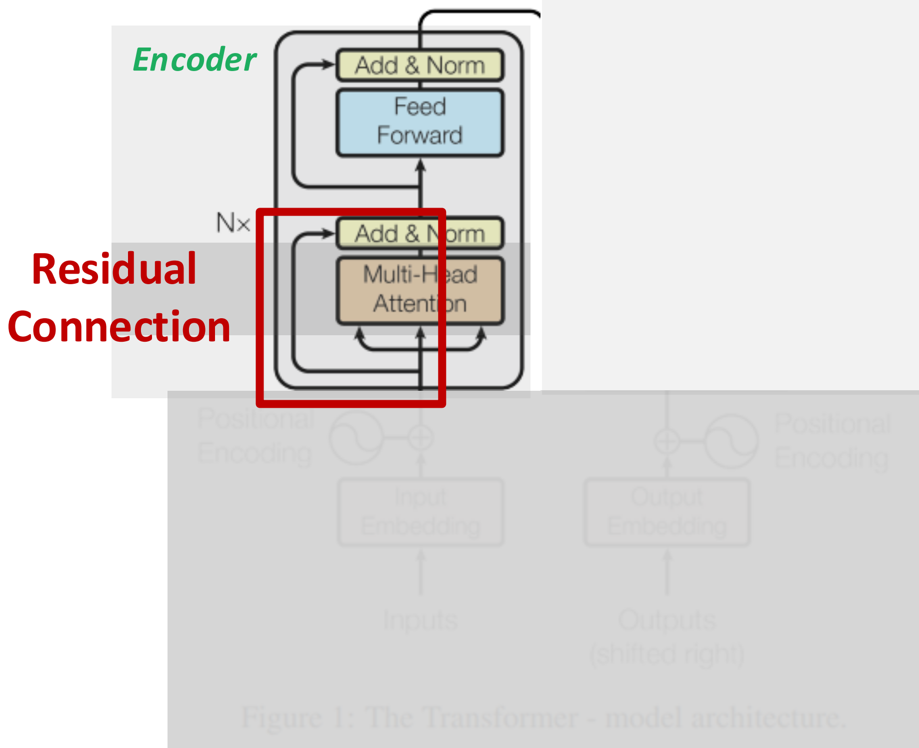
An example of the **attention mechanism** following long-distance dependencies **in the encoder self-attention in layer 5 of 6**.

Many of the attention heads attend to a distant dependency of the verb ‘making’, completing the phrase ‘**making…more difficult**’. **Attentions here shown only for the word ‘making’.**

Different colours represent **different heads**.

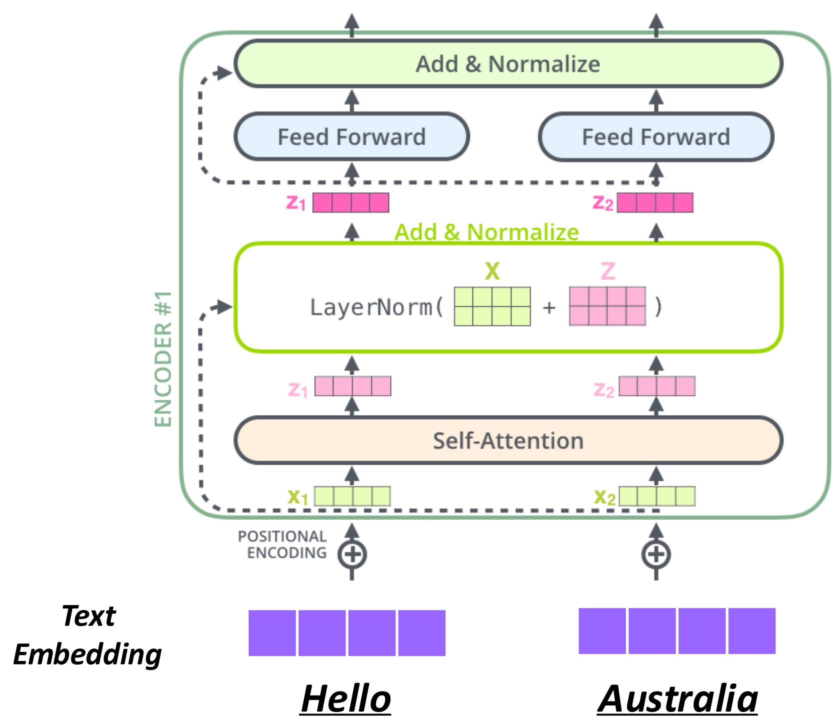
Transformer: Main Points and Decoder

Transformer – Efficient and Faster!



Residual Connection

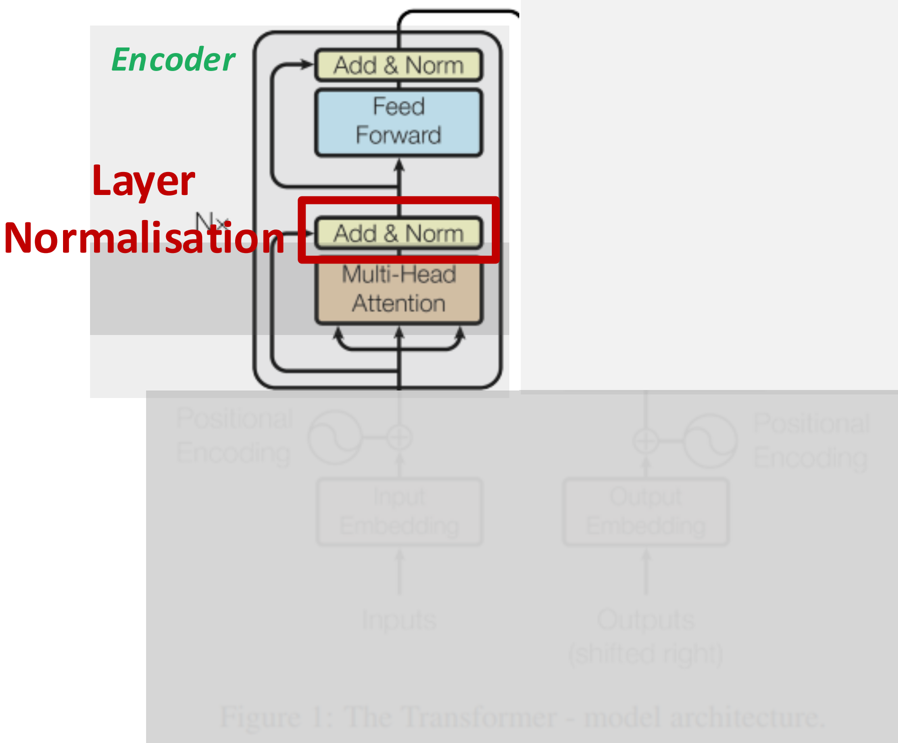
A trick to help models train better

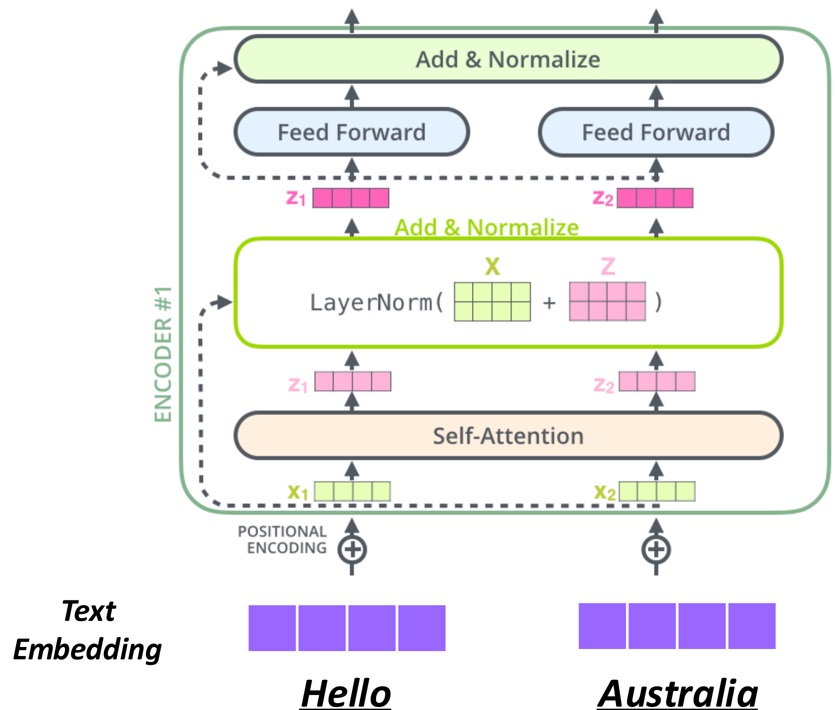


Transformer – Efficient and Faster!

Layer Normalisation (Normalising Gradient)

cut down on uninformative variation in hidden vector values by normalizing to unit mean and standard deviation within each layer.





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Decoder-Only Block

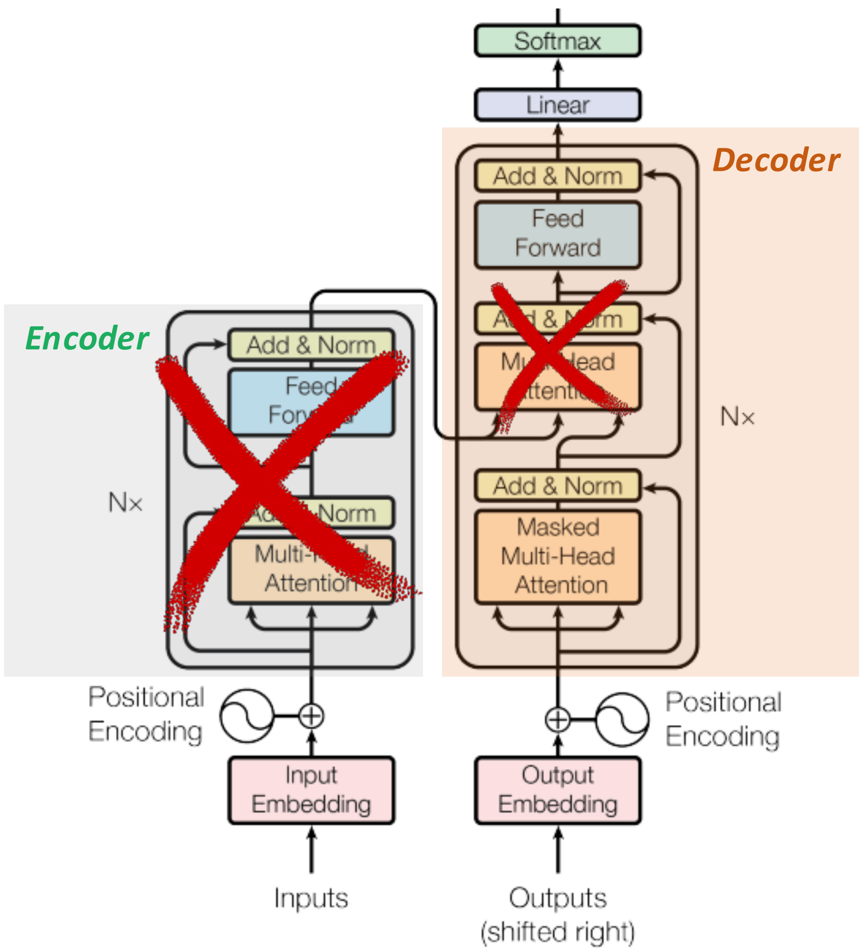
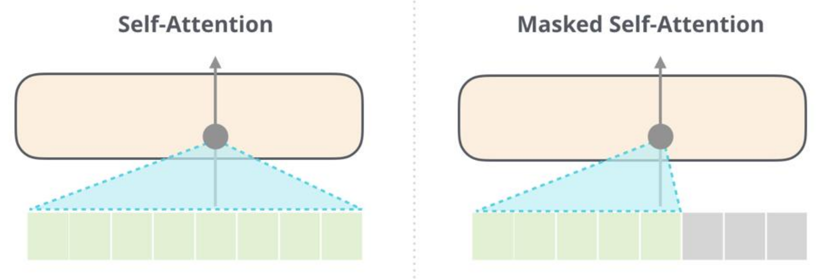


Figure 1: The Transformer - model architecture.



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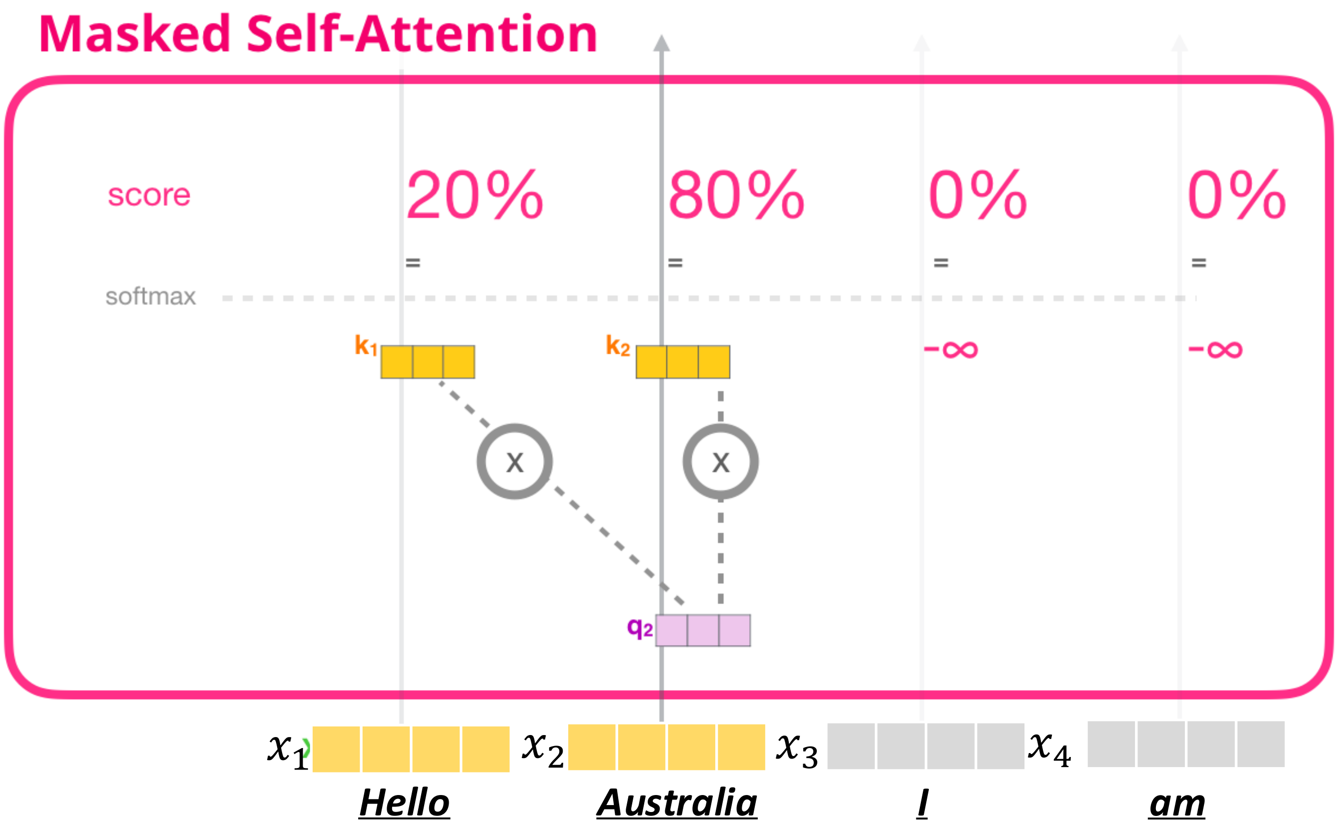

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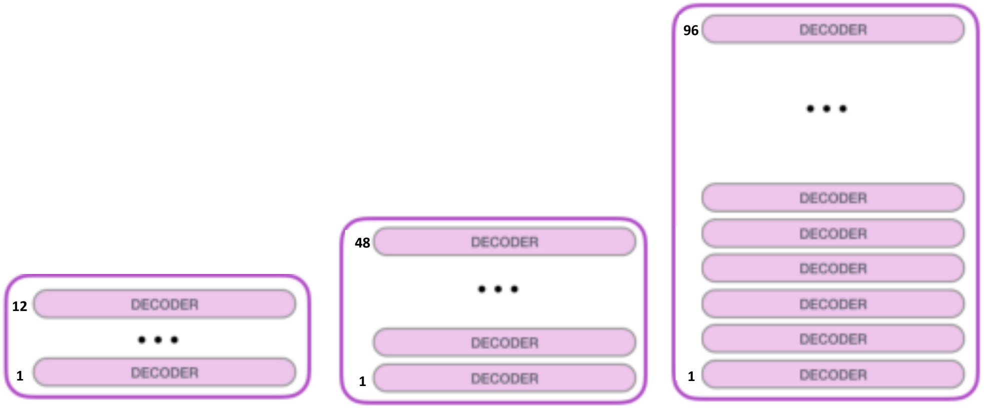
3 Generative Pretrained Transformers (GPT)



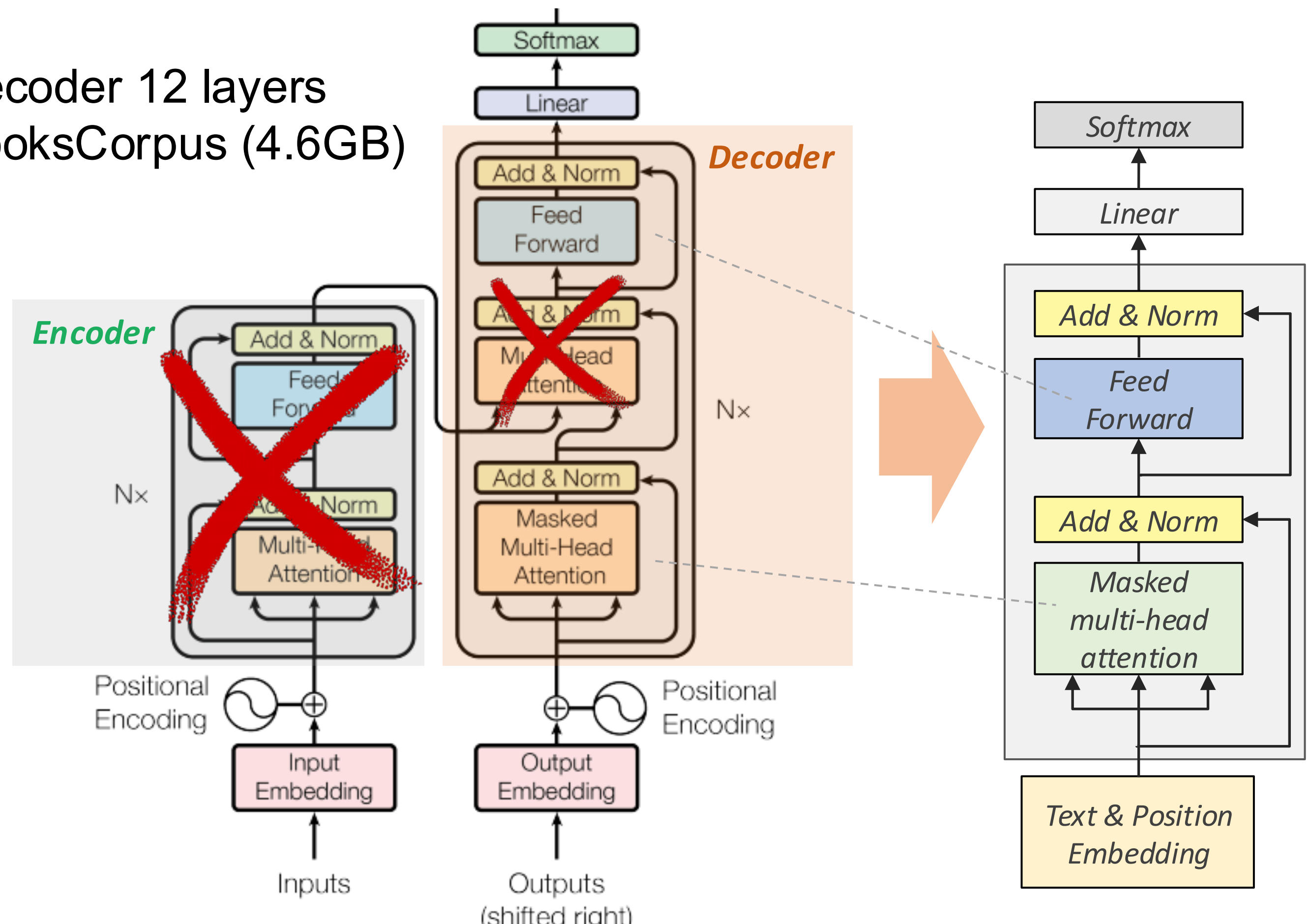
	GPT-1	GPT-2	GPT-3
Training Data	BookCrawl	WebText	CommonCrawl
Parameters	117M	1.5B	175B
Layers	12	48	96
Dimensions	768	1600	12288

Unsupervised Learning

Generative Pretrained Transformers (GPT)

GPT-1 (117M parameters)

- Decoder 12 layers
- BooksCorpus (4.6GB)



GPT-1

Figure 1: The Transformer - model architecture.

GPT-2 (1.5B parameters)

- Decoder 12 - 48 layers
- More Data (40GB)

Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

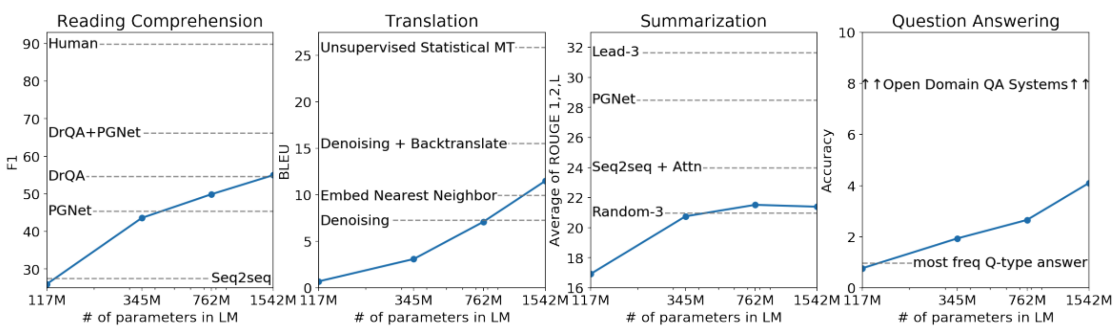
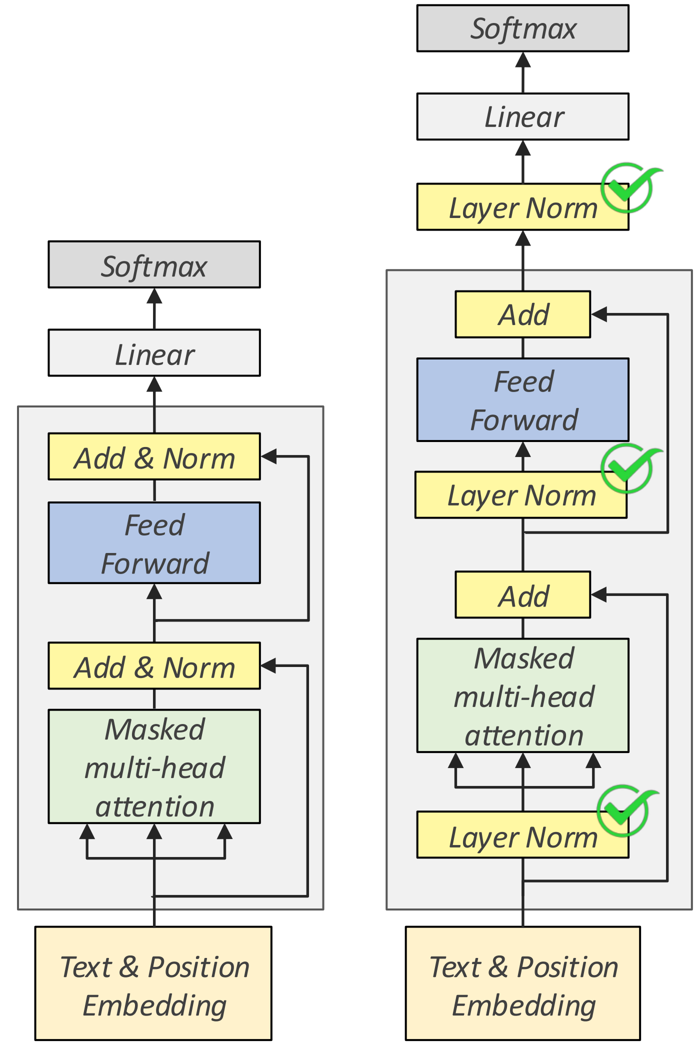


Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.



GPT-1 ***GPT-2***

GPT-2 (1.5B parameters)

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

Table 5. The 30 most confident answers generated by GPT-2 on the development set of Natural Questions sorted by their probability according to GPT-2. None of these questions appear in WebText according to the procedure described in Section 4.

GPT-2 (1.5B parameters)

- Emergent ability in GPT-2 is zero-shot learning
  - the ability to do many tasks with no examples, and no gradient updates

- Beats SoTA on language modelling benchmarks
  with no task-specific fine-tuning

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	**21.8**
117M	**35.13**	45.99	**87.65**	**83.4**	**29.41**	65.85	1.16	1.17	37.50	75.20
345M	**15.60**	55.48	**92.35**	**87.1**	**22.76**	47.33	1.01	**1.06**	26.37	55.72
762M	**10.87**	**60.12**	**93.45**	**88.0**	**19.93**	**40.31**	**0.97**	**1.02**	22.05	44.575
1542M	**8.63**	**63.24**	**93.30**	**89.05**	**18.34**	**35.76**	**0.93**	**0.98**	**17.48**	42.16

Table 3. Zero-shot results on many datasets. No training or fine-tuning was performed for any of these results. PTB and WikiText-2 results are from (Gong et al., 2018). CBT results are from (Bajgar et al., 2016). LAMBADA accuracy result is from (Hoang et al., 2018) and LAMBADA perplexity result is from (Grave et al., 2016). Other results are from (Dai et al., 2019).

Generative Pretrained Transformers (GPT)

GPT-2 (1.5B parameters)

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	LAMBADA (PPL)
SOTA	99.8
117M	**35.13**
345M	**15.60**
762M	**10.87**
1542M	**8.63**

(1) *Context:* "Yes, I thought I was going to lose the baby." "I was scared too," he stated, sincerity flooding his eyes. "You were ?" "Yes, of course. Why do you even ask?" "This baby wasn't exactly planned for."
Target sentence: "Do you honestly think that I would want you to have a _____ ?"
Target word: miscarriage

(2) *Context:* "Why?" "I would have thought you'd find him rather dry," she said. "I don't know about that," said Gabriel. "He was a great craftsman," said Heather. "That he was," said Flannery.
Target sentence: "And Polish, to boot," said _____.
Target word: Gabriel

(3) *Context:* Preston had been the last person to wear those chains, and I knew what I'd see and feel if they were slipped onto my skin-the Reaper's unending hatred of me. I'd felt enough of that emotion already in the amphitheater. I didn't want to feel anymore. "Don't put those on me," I whispered. "Please."
Target sentence: Sergei looked at me, surprised by my low, raspy please, but he put down the _____.
Target word: chains

(4) *Context:* They tuned, discussed for a moment, then struck up a lively jig. Everyone joined in, turning the courtyard into an even more chaotic scene, people now dancing in circles, swinging and spinning in circles, everyone making up their own dance steps. I felt my feet tapping, my body wanting to move.
Target sentence: Aside from writing, I 've always loved _____.
Target word: dancing

GPT and LLM capabilities

GPT-3 (175B parameters)

- Data (40GB → over 600GB)

Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

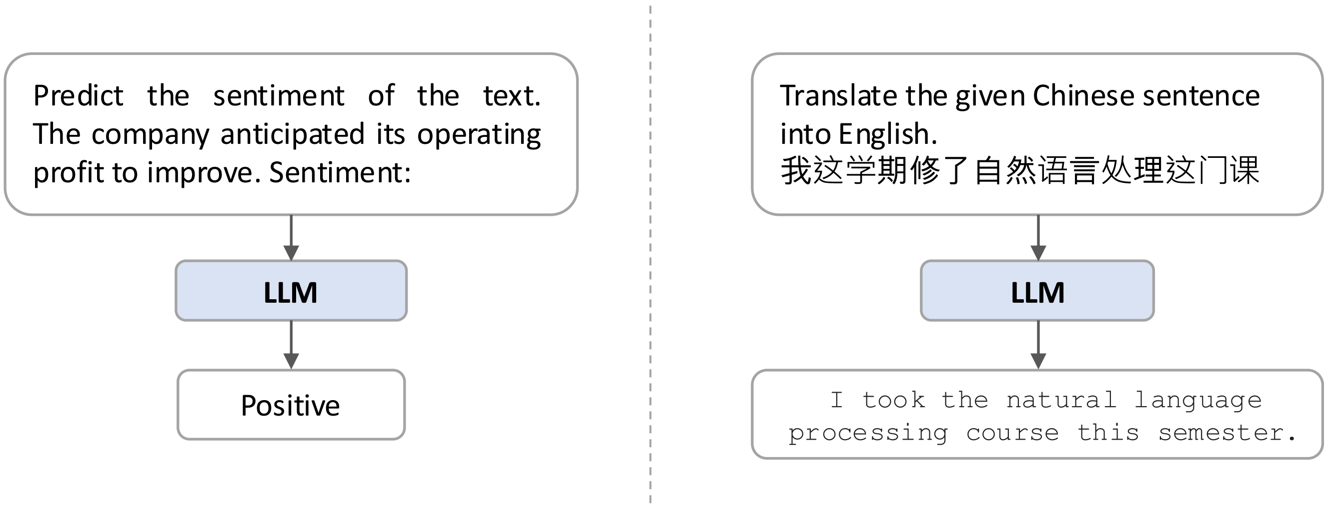
Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

GPT-3 with Instruction Following

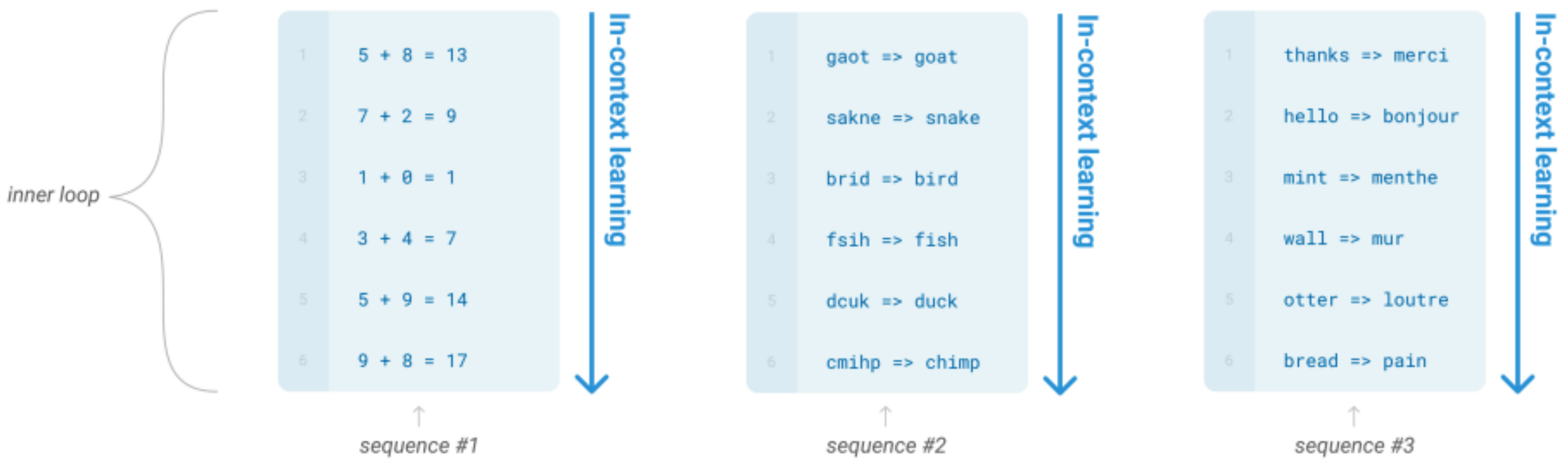
Models of a certain scale (175B) seem to begin to understand instructions without any demonstrations (zero-shot)



3 GPT and LLM capabilities

GPT-3 (175B parameters)

- Specify a task by prepending examples of the tasks before your example
- Called 'in-context learning', to stress that no gradient updates are performed when learning a new task



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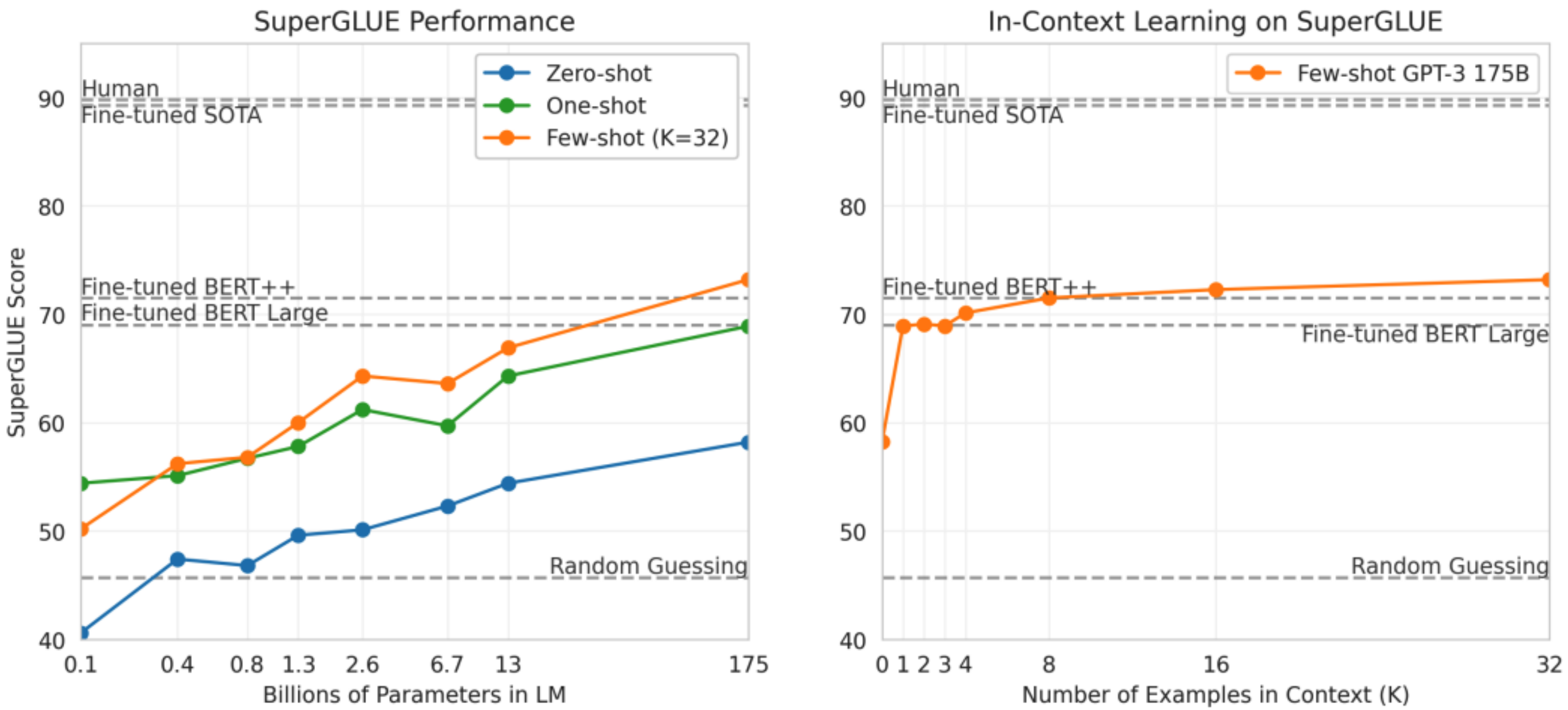

3 GPT and LLM capabilities

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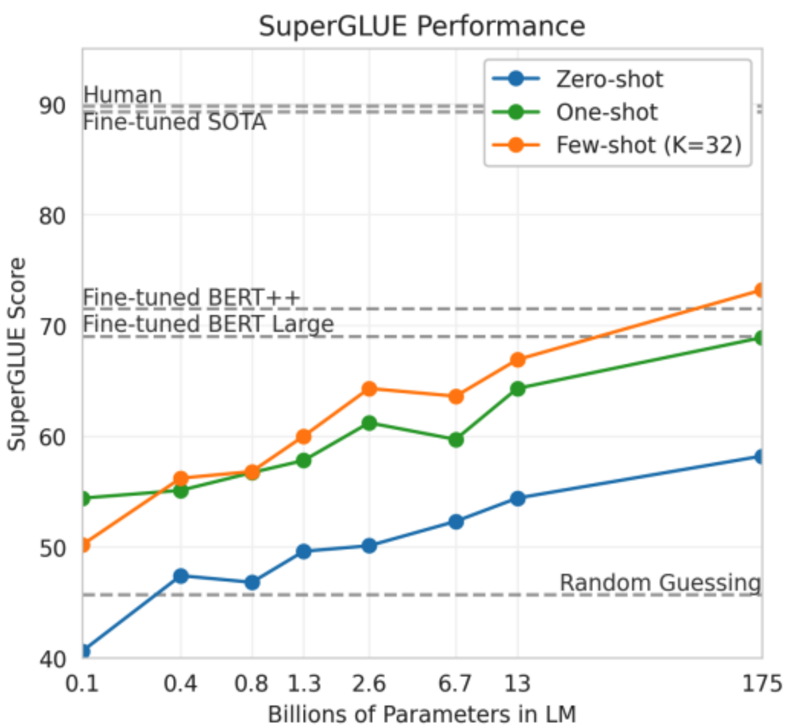
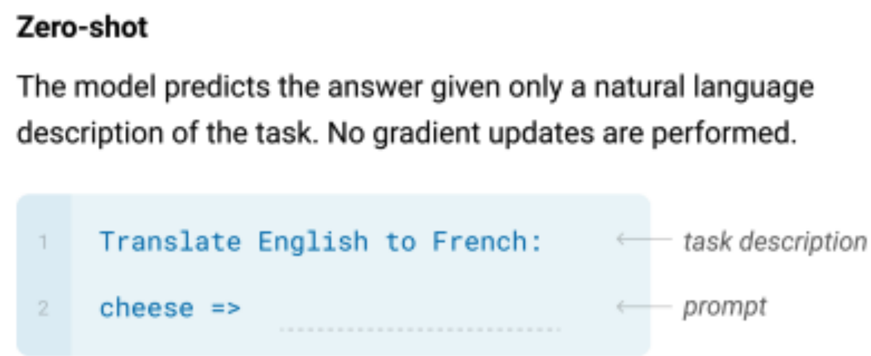
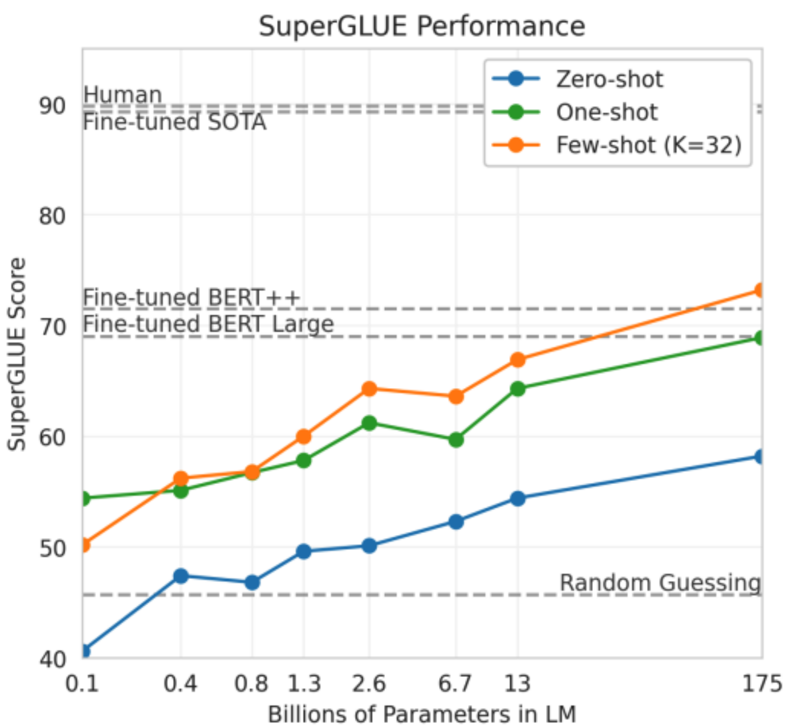


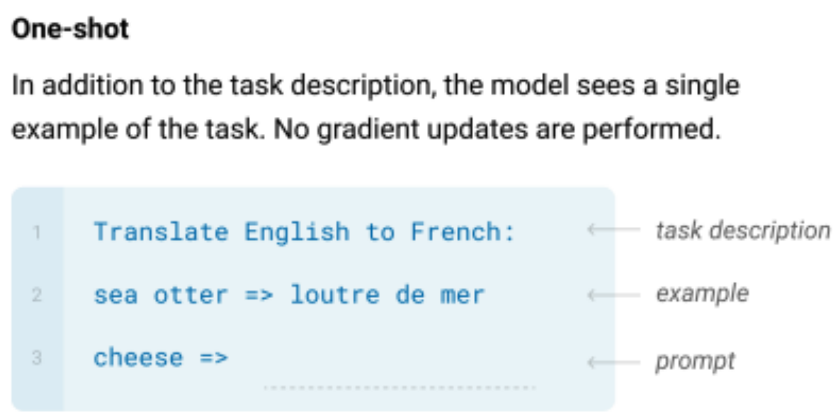


3 GPT and LLM capabilities

GPT-3 (175B parameters)

- Specify a task by prepending examples of the tasks before your example
- Called ‘in-context learning’, to stress that no gradient updates are performed when learning a new task





3 GPT and LLM capabilities

GPT-3 (175B parameters)

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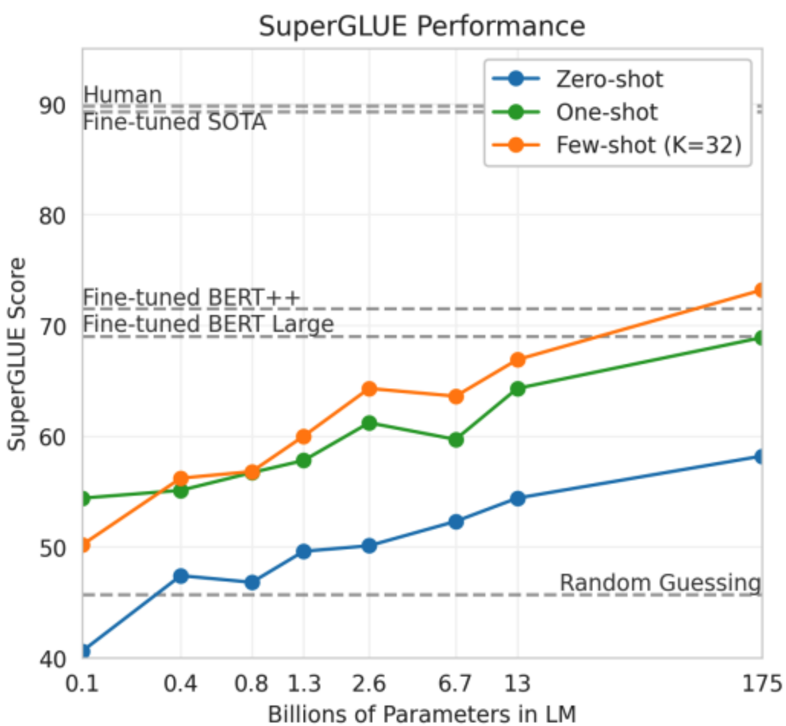

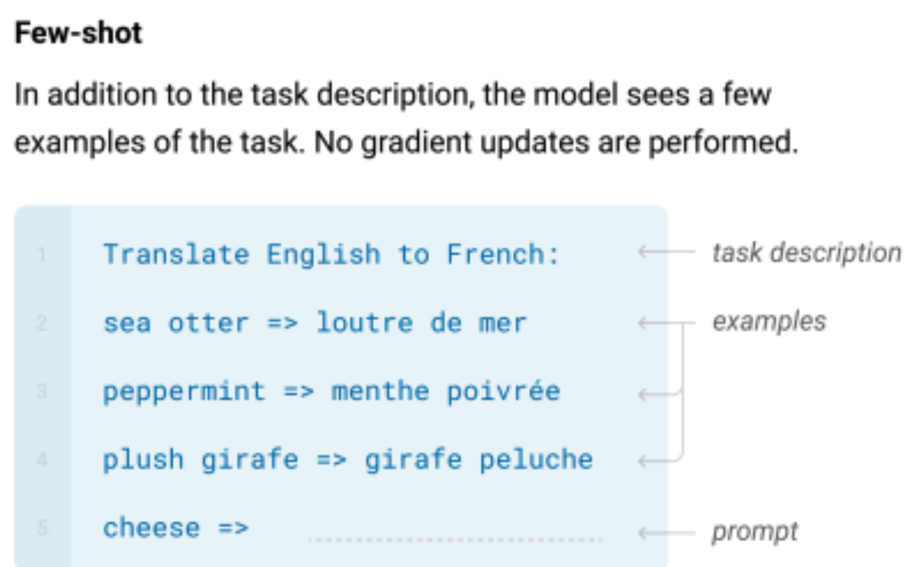

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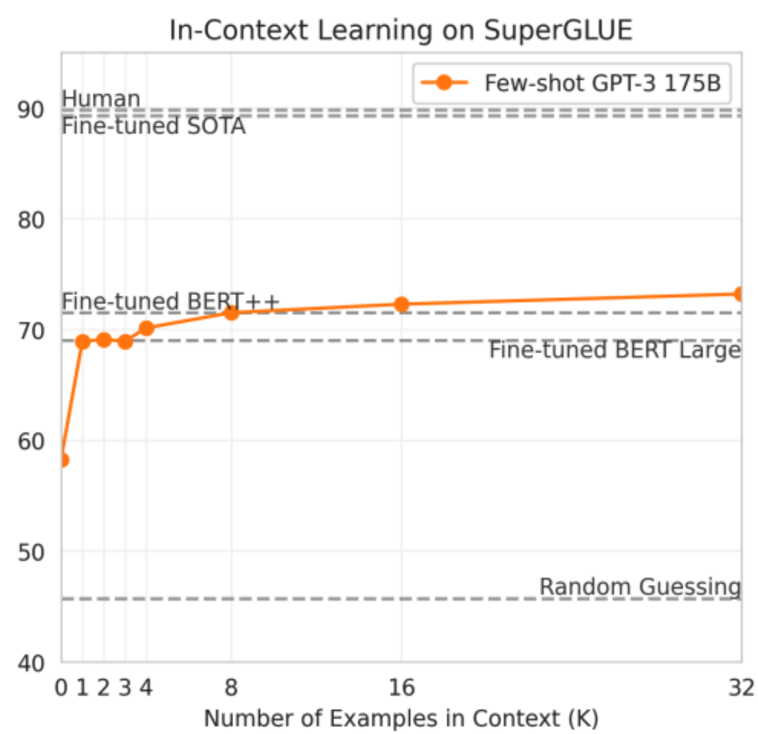
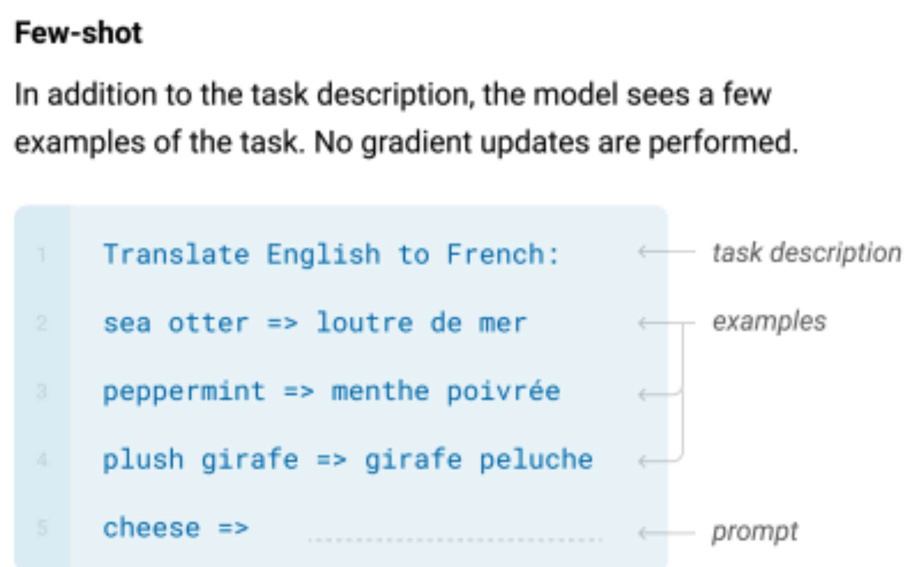
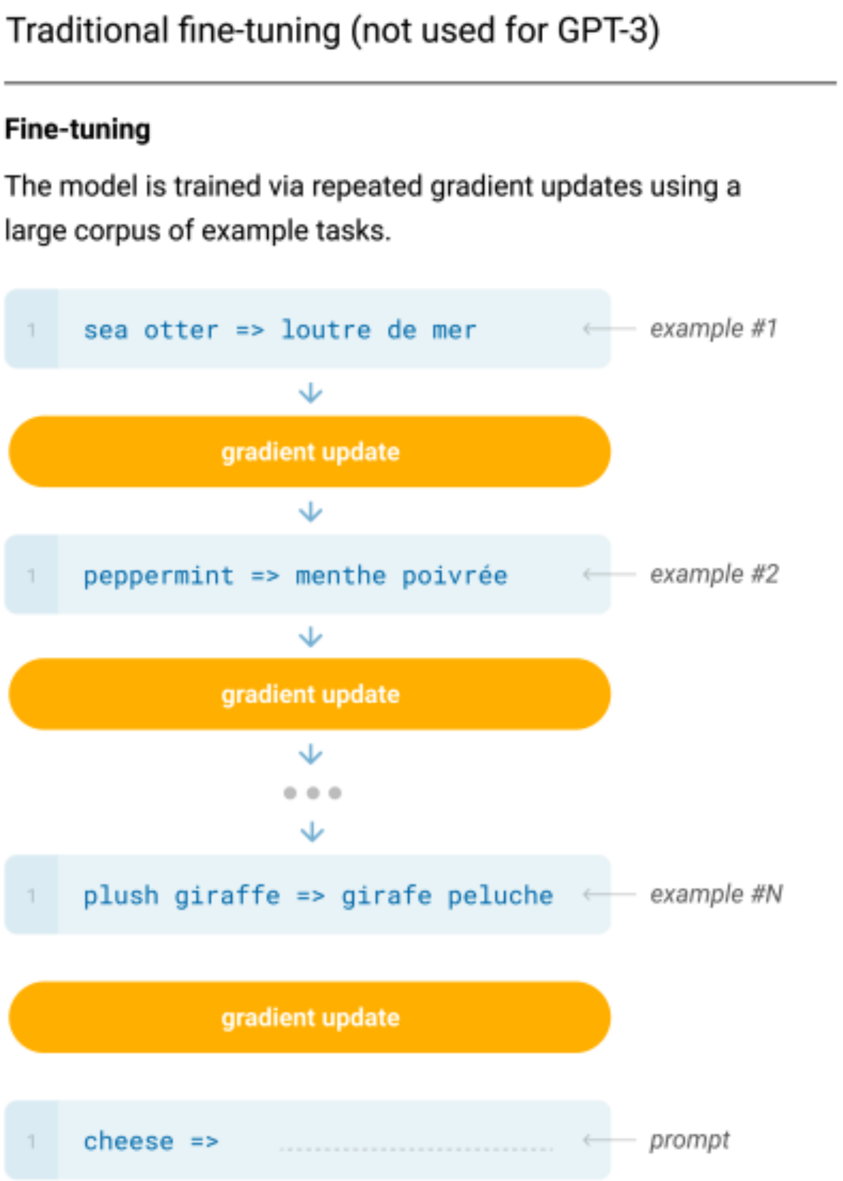




3 GPT and LLM capabilities

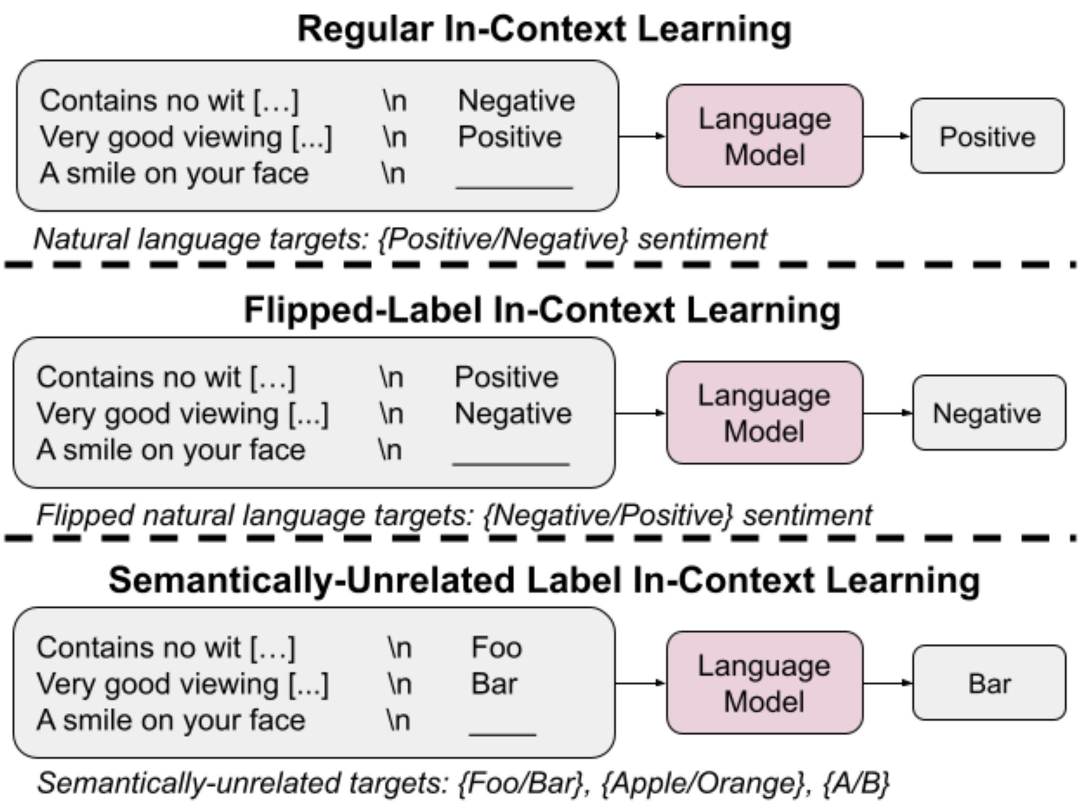
GPT-3 (175B parameters)

- What about the traditional fine-tuning methods?



In-context Learning

Solve a new task by showing a handful of expected input-output (aka demonstrations), **without any fine-tuning**



Limitations…

What if the task is too difficult? so that LLMs to learn through prompting alone.

- (especially, richer and multi-step reasoning tasks)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

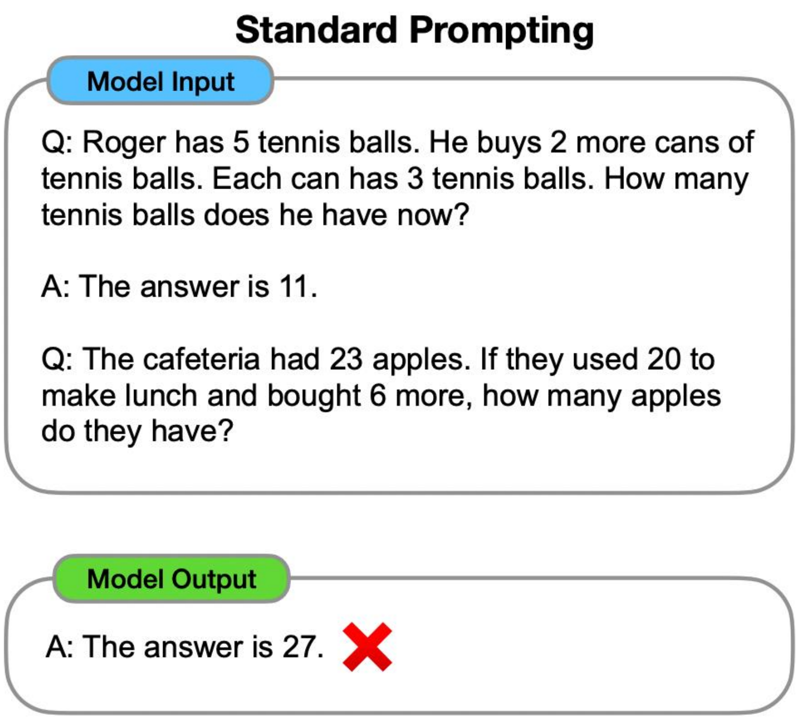
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

3 GPT and LLM capabilities

Limitations…

What if the task is too difficult? so that LLMs to learn through prompting alone.

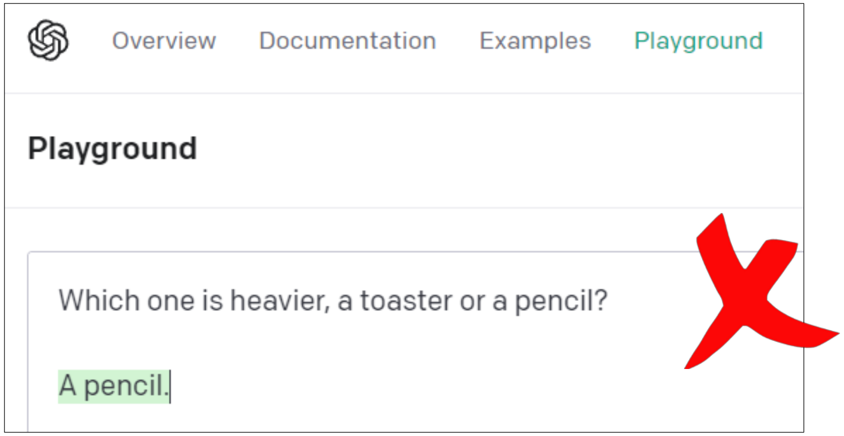
- (especially, richer and multi-step reasoning tasks)

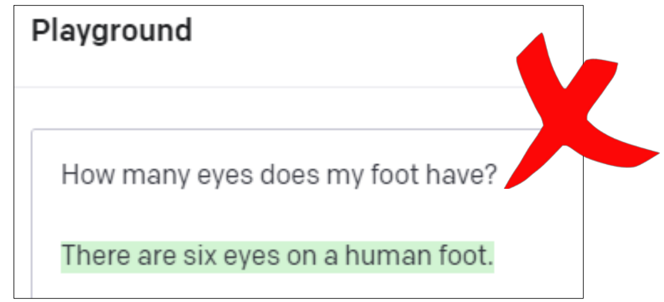


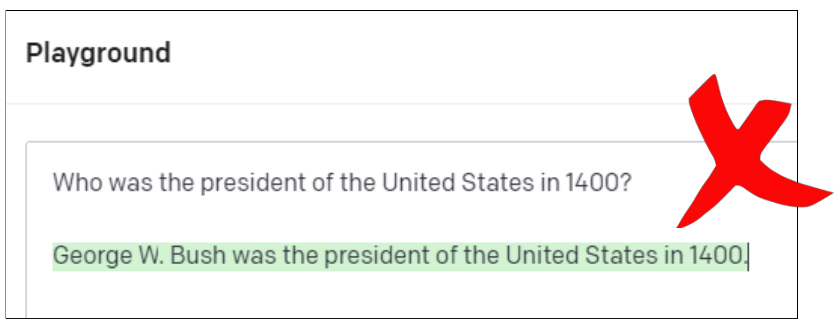
Alternative Solution:
Change the prompting style!
(Learn more in **Lecture 13, 14**)

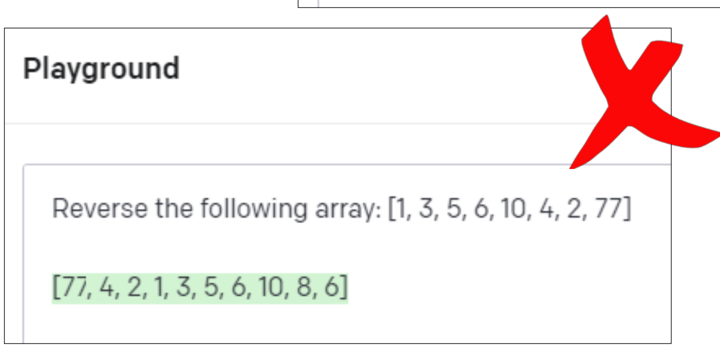
3 GPT and LLM capabilities

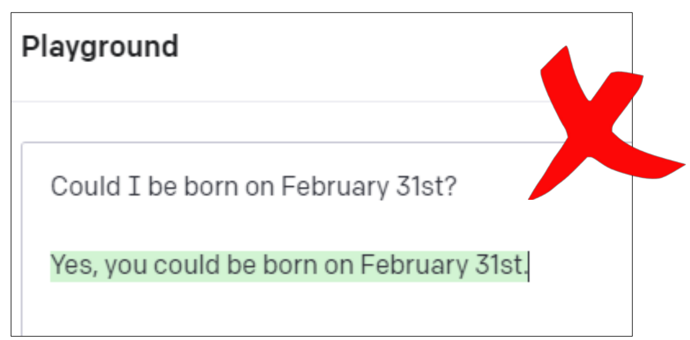
And GPT-3 is not perfect as it does not do the reasoning or checking the answer.



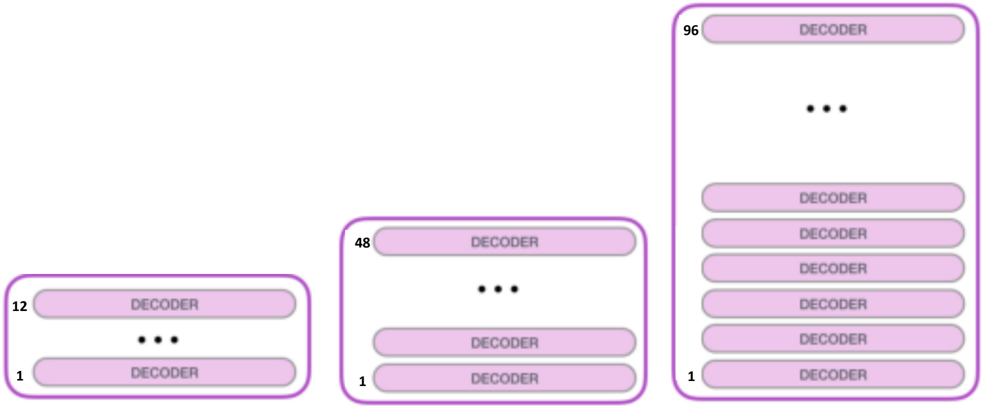








3 GPT and RLHF



	GPT-1	GPT-2	GPT-3
Training Data	BookCrawl	WebText	CommonCrawl
Parameters	117M	1.5B	175B
Layers	12	48	96
Dimensions	768	1600	12288

ChatGPT

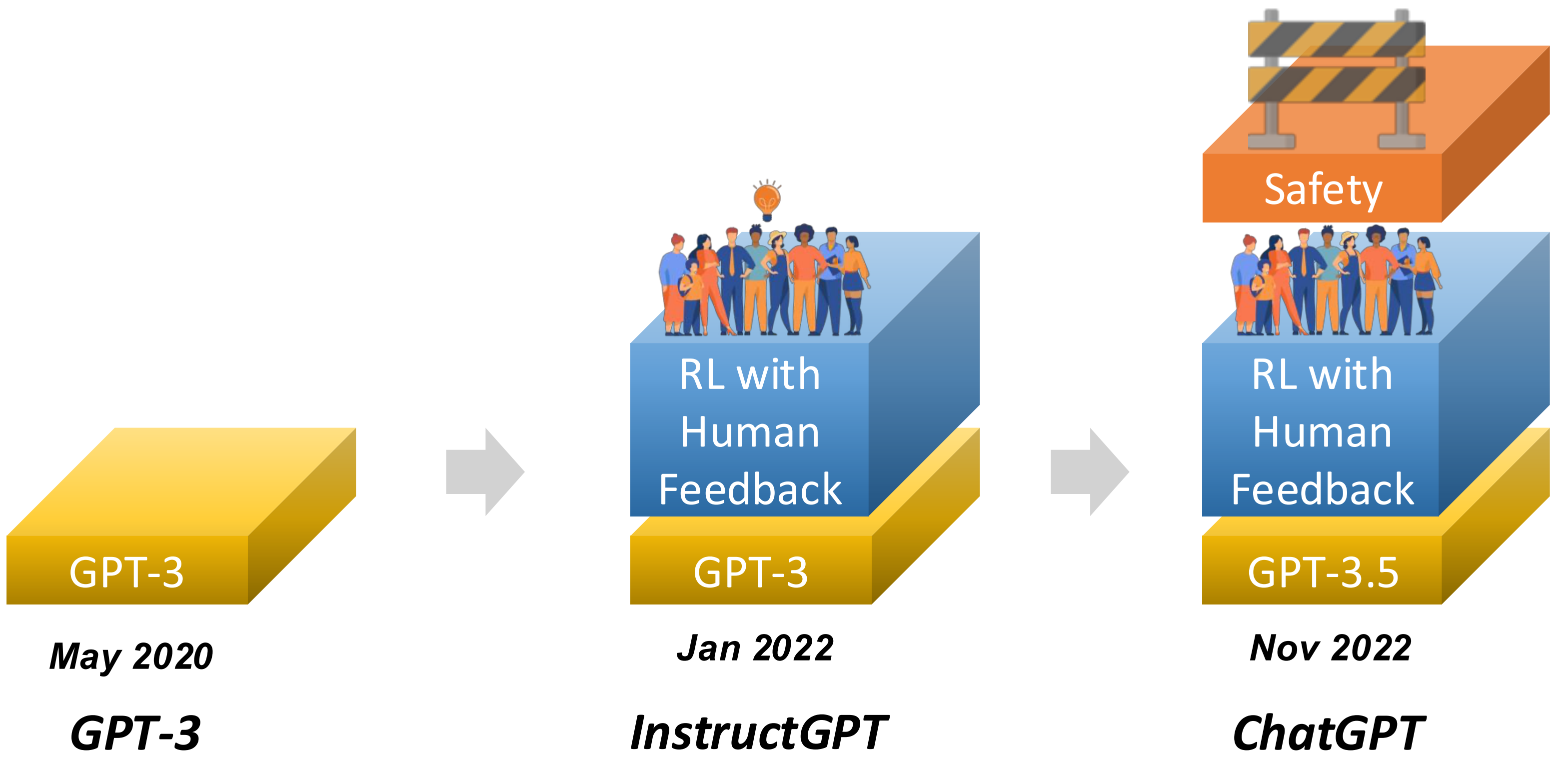
+ GPT-3.5

Supervised

Reinforcement Learning

Unsupervised Learning

3 GPT and RLHF



GPT3 vs InstructGPT

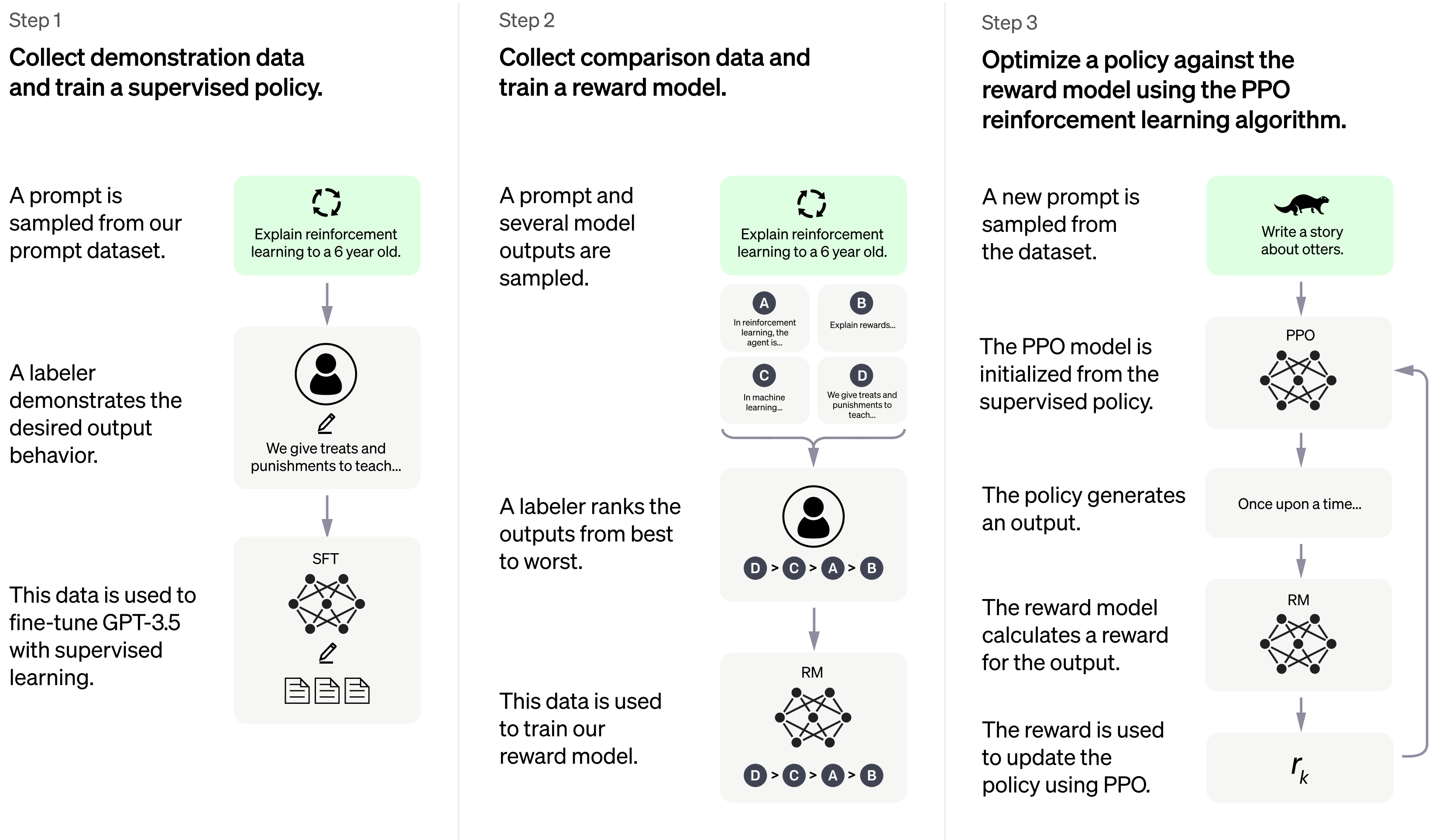
From GPT-3

> ***Write a short story about the moon and the stars:*** *Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars? Does the moon like the stars?*

From InstructGPT

> ***Write a short story about the moon and the stars:***
>
> *Once upon a time, the moon and the stars lived together in the sky. They were the best of friends, and they loved spending their time together. The moon would light up the sky at night, and the stars would twinkle and shine. They were always happy and they never fought.*
>
> *But one day, the moon and the stars began to grow apart. The moon started to feel left out and the stars began to ignore her. The moon didn't understand what was happening, and she felt very lonely.*
>
> *One night, the moon decided to take a walk through the sky. She passed by all the other planets and stars, but they all just ignored her. The moon began to cry and she felt more alone than ever.*
>
> *Suddenly, the moon heard a voice. It was the sun. He had been watching the moon from a distance and he had seen how sad she was. The sun told the moon that he loved her and that he would always be there for her. The moon was so happy to hear this, and she felt much better.*
>
> *The moon and the sun went back to the sky and they were finally able to be happy again.*

3 GPT and RLHF



More detailed RLHF procedure will be learned in Lecture 17 and 18

It is good that the AI still requires the human power.

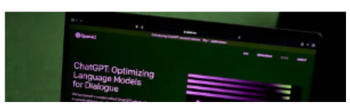
OpenAI is hiring developers to make ChatGPT better at coding

Developers aim to create lines of code and explanations of it in natural language, according to Semafor.

OpenAI is hiring hundreds of contractors from different parts of the globe to help ChatGPT get better at coding, according to a report from Semafor.

More specifically, the company is hiring computer programmers to create training data that will include lines of code, as well as explanations of the code written in natural language.

/ ZDNET recommends



OPENAI IS HIRING AN

Expert AI Teacher (Contract)

Contractor
Remote

About the Team

The Human Data Team is focused on empowering AI teachers to train OpenAI's models to solve complex problems in order to provide value to humanity. We believe that Expert AI teachers are the key to unlocking the potential of our models to be as accurate and helpful as experts.

We are building a system that uses experts' guidance to teach our models how to understand difficult questions and execute complex instructions. This will foster collaboration between AI and humans that aligns with our shared vision and values.